

VASTRA

Supabase Database Architecture & Complete Schema Dictionary

How the four Vastra applications share one secure PostgreSQL data platform

Version 1.0

Prepared 11 July 2026

Supabase/PostgreSQL implementation blueprint

Customer App • Merchant App • Captain App • Admin & Operations

How to use this document

Purpose

This is the canonical logical data model for Vastra. It explains where every data domain belongs, which application may access it, how critical operations should be executed, and the exact attributes expected for 103 proposed tables. The separate Runbook PDF contains the manual setup sequence, keys, commands, and SQL migrations.

- 1. Use the architecture chapters to decide application/backend boundaries before coding screens.
- 2. Use the table dictionary when writing repositories, TypeScript types, DTOs, RLS tests and admin reports.
- 3. Apply only the MVP tables required by the current sprint; later AI and advanced operations tables are already reserved to avoid redesign.
- 4. Treat the generated SQL pack as an implementation starter: run it locally, review it, and test every RLS policy before production.

Document map

Item	Details
Part I	Architecture, Supabase components, conventions and access boundaries
Part II	Relationships and business transactions across Customer, Merchant, Captain and Admin
Part III	Complete table and attribute dictionary by module
Part IV	RLS, Storage, Realtime, AI data, performance, backups and rollout
Companion files	Manual Setup & SQL Runbook PDF plus the downloadable SQL/environment ZIP

1. Recommended Supabase architecture

Supabase is the managed PostgreSQL foundation, not the entire business backend. The apps may use Auth, safe reads, Realtime subscriptions and permitted uploads directly, while critical money, inventory, role, order and delivery mutations pass through trusted backend transactions or narrowly reviewed database functions.

Item	Details
Supabase Auth	Identity, OTP/social login, JWT sessions. Application profile data stays in public.profiles.
PostgreSQL	Authoritative relational data, constraints, transactions, PostGIS and pgvector.
Row Level Security	Row-by-row protection for all tables exposed through the Data API.
Supabase Storage	Product/shop media plus private KYC, return evidence, body scans and try-on outputs.
Supabase Realtime	Foreground order status, merchant alerts, delivery events, support chat and notifications.
Edge Functions / Node backend	Payment webhooks, inventory/order transactions, dispatch, alerts, payouts and AI orchestration.
External services	FCM/APNs, payment gateway, maps/routing, SMS/WhatsApp and optional AI providers.

Non-negotiable boundary

A customer, merchant or captain client must never directly set payment status, reduce arbitrary inventory, assign itself a delivery, approve a refund, alter a settlement, promote an account

to ADMIN, or bypass an order state transition.

2. Core design conventions

Item	Details
Identity	Supabase owns auth.users. public.profiles uses the same UUID and stores the Vastra account type and app profile.
Primary keys	UUID with gen_random_uuid() for business records; bigint identity for append-only high-volume logs.
Money	BIGINT paise. ₹1,499.50 is stored as 149950, avoiding floating-point errors.
Time	TIMESTAMPTZ in UTC; apps render in the user/shop timezone.
Location	PostGIS geography(Point,4326) for points and geography(Polygon,4326) for zones.
Media	Database stores Storage object paths and metadata, not image bytes or permanent signed URLs.
Inventory	Always variant-level. Available = stock_on_hand - reserved_quantity - damaged_quantity.
History	Inventory movements, order/return statuses, delivery events, payment events and audit logs are append-only.
Deletion	Soft-delete public business entities; retention/delete private body images and KYC data under explicit policies.
Concurrency	Lock inventory rows and use database transactions or optimistic version columns.
Naming	snake_case tables/columns; stable machine-readable status/action codes; customer-facing numbers separate from UUIDs.
Deployment	Versioned migrations only. Avoid manual remote schema edits after migration workflow begins.

3. Four-application access model

Item	Details
Customer App	Public catalogue; own profile, addresses, favourites, groups, carts, orders, payments, returns, fit and try-on data.
Merchant App	Own shop, catalogue, variants, barcode/photo inventory, offline sales, incoming orders, packing, returns, offers and settlements.
Captain App	Own captain profile/documents; offered/assigned delivery tasks, live location, events, COD, earnings and payouts.
Admin Platform	Permission-controlled operations, support, KYC, moderation, finance, campaigns, field visits, risk and audit.
Trusted backend	Cross-user transactions, secret-key operations, payment verification, dispatch, alert retries, refunds and payouts.

4. Relationship overview

Entity flow

auth.users -> profiles -> customer_profiles / merchant_profiles / captain_profiles / admin_profiles
 merchant_profiles -> shops -> products -> product_variants -> inventory_balances
 customer_profiles -> carts -> orders -> order_items -> delivery_tasks -> captain_profiles

orders -> payments / refunds / return_requests / merchant_settlements
 shops <-> customer_profiles through shop_favourites
 style_groups -> members -> themes / polls / recommendations / moodboards
 customer_events -> recommendation_sets -> recommendation_items
 customer_body_profiles -> measurements -> fit_recommendations / body_scan_sessions / virtual_tryon_sessions
 admin roles -> permissions; support / approval / fraud / audit surround every domain

5. Critical transaction patterns

Item	Details
Place order	Validate current prices; lock all inventory rows; verify available stock; create order/items; create reservations; increase reserved quantities; create payment reference; commit atomically.
Merchant accepts	Check order is WAITING_FOR_MERCHANT; confirm reservations; save preparation time; append status history; create dispatch task; notify customer and captains.
Merchant cannot fulfil	Record affected item/reason/evidence; release reservations; correct stock/damaged balance if necessary; cancel or start replacement flow; initiate refund when captured.
Offline sale	Lock each variant balance; verify availability; create sale/items; decrement stock_on_hand; append movements; publish updated availability.
Captain pickup	Verify assignment, geofence and pickup code; append delivery event; change task/order status; start customer tracking.
Delivery completion	Verify OTP/proof and COD; append delivery event/status history; calculate captain earning; mark settlement eligibility.
Return/refund	Check eligibility; create return and pickup task; inspect item; make inventory decision; create approval/refund; process gateway idempotently.
Role or bank change	Use maker-checker approval, merchant/user OTP where required, and append audit log without exposing sensitive values.

6. Module summary

Module	Scope
Identity & Access	12 tables
Marketplace	4 tables
Catalogue	7 tables
Inventory	4 tables
Commerce	10 tables
Payments	3 tables
Returns	4 tables
Finance	4 tables

Module	Scope
Logistics	8 tables
Group Style	9 tables
Engagement	3 tables
Notifications	3 tables
Promotions	3 tables
Campaigns	2 tables
Intelligence	16 tables
Operations	2 tables
Field Operations	3 tables
Moderation	1 tables
Risk	1 tables
Governance	2 tables
Configuration	2 tables

Part III - Complete table and attribute dictionary

Identity & Access

12 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

profiles [MVP]

Application profile linked one-to-one with Supabase-managed auth.users. Self-sign-up always starts as CUSTOMER; trusted workflows promote merchant, captain or admin accounts.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	-	PK/FK	auth.users.id	Same UUID as the Supabase Auth user
account_type	text	No	'CUSTOMER'	CHECK	-	CUSTOMER, MERCHANT, CAPTAIN or ADMIN
full_name	text	Yes	-	-	-	Display and legal-facing name where appropriate
phone_number	text	Yes	-	-	-	Normalized E.164 phone copied from Auth for search/display
email	citext	Yes	-	-	-	Email copied from Auth when available
avatar_path	text	Yes	-	-	-	Storage object path, not a permanent public URL
status	text	No	'ACTIVE'	CHECK	-	ACTIVE, PENDING, BLOCKED, SUSPENDED or DELETED
preferred_language	text	No	'en'	-	-	UI language code
last_login_at	timestamptz	Yes	-	-	-	Last successful authentication timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Last update timestamp
deleted_at	timestamptz	Yes	-	-	-	Soft deletion marker

user_devices [MVP]

Registered mobile/web devices used for session visibility, push tokens, merchant ringing alerts and security controls.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Device registration ID
user_id	uuid	No	-	FK	profiles.id	Owner of the device
device_fingerprint	text	No	-	UNIQUE-PART	-	App-generated stable installation identifier
platform	text	No	-	CHECK	-	ANDROID, IOS or WEB
push_provider	text	Yes	-	CHECK	-	FCM, APNS or WEB_PUSH
push_token	text	Yes	-	-	-	Current push token; rotate on refresh
app_name	text	No	-	CHECK	-	CUSTOMER, MERCHANT, CAPTAIN or ADMIN
app_version	text	Yes	-	-	-	Installed application version
device_model	text	Yes	-	-	-	Device model for support diagnostics
os_version	text	Yes	-	-	-	Operating system version
notification_enabled	boolean	No	true	-	-	Whether OS/app notifications are enabled

Attribute	Type	Null?	Default	Key	Reference	Purpose
order_sound_enabled	boolean	No	true	-	-	Merchant new-order ringtone preference
last_seen_at	timestamptz	Yes	-	-	-	Last foreground or authenticated activity
revoked_at	timestamptz	Yes	-	-	-	Revocation timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

Additional constraints: unique (user_id, device_fingerprint)

addresses [MVP]

Reusable customer delivery addresses and contact snapshots. Shop addresses are stored directly on shops to avoid ownership ambiguity.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Address ID
user_id	uuid	No	-	FK	profiles.id	Address owner
label	text	Yes	-	-	-	Home, Work, Other, etc.
recipient_name	text	No	-	-	-	Recipient name
phone_number	text	No	-	-	-	Contact phone for delivery
line1	text	No	-	-	-	Building/door/street
line2	text	Yes	-	-	-	Additional address details
landmark	text	Yes	-	-	-	Nearby landmark
area	text	No	-	-	-	Locality/area
city	text	No	-	-	-	City
state	text	No	-	-	-	State
postal_code	text	No	-	-	-	PIN/postal code
country_code	text	No	'IN'	-	-	ISO country code
location	geography(Point, 4326)	Yes	-	-	-	Precise map location
is_default	boolean	No	false	-	-	Default address flag
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

customer_profiles [MVP]

Customer-specific data that should not be mixed into the generic profile.

Attribute	Type	Null?	Default	Key	Reference	Purpose
user_id	uuid	No	-	PK/FK	profiles.id	Customer profile owner
date_of_birth	date	Yes	-	-	-	Optional age-related personalization input
gender_preference	text	Yes	-	-	-	Catalogue preference, not identity verification
default_address_id	uuid	Yes	-	FK	addresses.id	Preferred delivery address
profile_completed	boolean	No	false	-	-	Onboarding completion marker
created_at	timestamptz	No	now()	-	-	Creation timestamp

Attribute	Type	Null?	Default	Key	Reference	Purpose
updated_at	timestamptz	No	now()	-	-	Update timestamp

customer_preferences [MVP]

Explicit fashion, budget, fit and notification preferences used by rule-based recommendations and later ML models.

Attribute	Type	Null?	Default	Key	Reference	Purpose
customer_id	uuid	No	-	PK/FK	customer_profiles.user_id	Customer
preferred_category_ids	uuid[]	No	'{}'	-	categories.id[]	Preferred categories
preferred_colours	text[]	No	'{}'	-	-	Preferred colours
avoided_colours	text[]	No	'{}'	-	-	Colours to suppress
preferred_styles	text[]	No	'{}'	-	-	Style tags
preferred_occasions	text[]	No	'{}'	-	-	Occasion tags
preferred_brands	text[]	No	'{}'	-	-	Brand preferences
min_budget_paise	bigint	Yes	-	CHECK	-	Minimum preferred price
max_budget_paise	bigint	Yes	-	CHECK	-	Maximum preferred price
fit_preference	text	No	'REGULAR'	CHECK	-	SLIM, REGULAR, RELAXED or OVERSIZED
include_footwear	boolean	No	true	-	-	Include footwear in outfit recommendations
include_accessories	boolean	No	true	-	-	Include belts, watches, bags and similar items
updated_at	timestamptz	No	now()	-	-	Update timestamp

merchant_profiles [MVP]

Merchant legal/onboarding profile. A single merchant login may own multiple shops later while MVP can enforce one active shop.

Attribute	Type	Null?	Default	Key	Reference	Purpose
user_id	uuid	No	-	PK/FK	profiles.id	Merchant account
legal_name	text	No	-	-	-	Legal owner/business name
business_type	text	Yes	-	-	-	Proprietorship, partnership, company, etc.
pan_last4	text	Yes	-	-	-	Masked PAN display only
pan_encrypted	text	Yes	-	-	-	Encrypted PAN value if stored
GST_number	text	Yes	-	-	-	GSTIN when applicable
onboarding_status	text	No	'STARTED'	CHECK	-	STARTED through ACTIVE/REJECTED/SUSPENDED
kyc_status	text	No	'PENDING'	CHECK	-	PENDING, IN_REVIEW, VERIFIED, REJECTED
approved_at	timestamptz	Yes	-	-	-	Approval timestamp
approved_by	uuid	Yes	-	FK	profiles.id	Trusted admin approver
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

captain_profiles [MVP]

Delivery partner profile, availability, vehicle and performance data.

Attribute	Type	Null?	Default	Key	Reference	Purpose
user_id	uuid	No	-	PK/FK	profiles.id	Captain account
captain_code	text	No	-	UNIQUE	-	Human-friendly captain identifier
kyc_status	text	No	'PENDING'	CHECK	-	PENDING, IN_REVIEW, VERIFIED, REJECTED
availability_status	text	No	'OFFLINE'	CHECK	-	OFFLINE, AVAILABLE, OFFERED, ASSIGNED, AT_PICKUP, DELIVERING, ON_BREAK, SUSPENDED
vehicle_type	text	Yes	-	CHECK	-	BIKE, SCOOTER, BICYCLE, WALKER, AUTO or CAR
vehicle_number	text	Yes	-	-	-	Registration number
driving_licence_last4	text	Yes	-	-	-	Masked licence number
driving_licence_encrypted	text	Yes	-	-	-	Encrypted licence number
rating_average	numeric(3,2)	No	0	CHECK	-	Cached rating average
rating_count	integer	No	0	CHECK	-	Rating count
completed_deliveries	integer	No	0	CHECK	-	Completed delivery count
cash_balance_paise	bigint	No	0	-	-	COD cash held/owed ledger summary
approved_at	timestamptz	Yes	-	-	-	Approval timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

admin_profiles [MVP]

Internal employee profile, department, city scope and second-factor status.

Attribute	Type	Null?	Default	Key	Reference	Purpose
user_id	uuid	No	-	PK/FK	profiles.id	Admin account
employee_code	text	No	-	UNIQUE	-	Internal employee identifier
department	text	No	-	-	-	Operations, support, finance, marketing, risk, etc.
city_scope	text[]	No	'{}'	-	-	Cities the admin can operate in
manager_id	uuid	Yes	-	FK	admin_profiles.user_id	Reporting manager
two_factor_enabled	boolean	No	false	-	-	MFA/2FA enforcement status
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

roles [MVP]

Named application roles for granular admin and operational access.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Role ID
code	text	No	-	UNIQUE	-	Stable machine-readable role code
name	text	No	-	-	-	Human-readable role name

Attribute	Type	Null?	Default	Key	Reference	Purpose
description	text	Yes	-	-	-	Scope and limitations
is_system_role	boolean	No	true	-	-	Protect seeded roles from accidental deletion
created_at	timestamptz	No	now()	-	-	Creation timestamp

permissions [MVP]

Fine-grained actions used by admin RLS helpers and backend authorization.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Permission ID
code	text	No	-	UNIQUE	-	Example order.reassign_captain
module	text	No	-	-	-	Functional module
name	text	No	-	-	-	Permission display name
description	text	Yes	-	-	-	Permission meaning
created_at	timestamptz	No	now()	-	-	Creation timestamp

user_roles [MVP]

Many-to-many mapping of users to internal roles; normally only admins receive these roles.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Assignment ID
user_id	uuid	No	-	FK	profiles.id	User receiving role
role_id	uuid	No	-	FK	roles.id	Assigned role
assigned_by	uuid	Yes	-	FK	profiles.id	Admin making assignment
assigned_at	timestamptz	No	now()	-	-	Assignment timestamp
revoked_at	timestamptz	Yes	-	-	-	Revocation marker

Additional constraints: unique (user_id, role_id)

role_permissions [MVP]

Many-to-many mapping between roles and allowed actions.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Mapping ID
role_id	uuid	No	-	FK	roles.id	Role
permission_id	uuid	No	-	FK	permissions.id	Permission
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (role_id, permission_id)

Marketplace

4 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

shops [MVP]

Canonical shop record shown in the customer marketplace and controlled by the approved merchant.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Shop ID
merchant_id	uuid	No	-	FK	merchant_profiles.user_id	Shop owner
shop_code	text	No	-	UNIQUE	-	Short code customers can search after installing Vastra
name	text	No	-	-	-	Public shop name
slug	text	No	-	UNIQUE	-	URL-safe shop identifier
description	text	Yes	-	-	-	Customer-facing description
phone_number	text	No	-	-	-	Shop contact number
email	citext	Yes	-	-	-	Shop email
address_line1	text	No	-	-	-	Shop address
address_line2	text	Yes	-	-	-	Additional address
landmark	text	Yes	-	-	-	Landmark
area	text	No	-	-	-	Locality
city	text	No	-	-	-	Operating city
state	text	No	-	-	-	State
postal_code	text	No	-	-	-	PIN code
country_code	text	No	'IN'	-	-	Country
location	geography(Point, 4326)	No	-	-	-	Geographic point for nearby discovery
logo_path	text	Yes	-	-	-	Storage object path
cover_image_path	text	Yes	-	-	-	Storage object path
verification_status	text	No	'PENDING'	CHECK	-	PENDING, IN_REVIEW, VERIFIED, REJECTED
operational_status	text	No	'CLOSED_FOR_DAY'	CHECK	-	OPEN, BUSY, TEMPORARILY_CLOSED, CLOSED_FOR_DAY, PAUSED, SUSPENDED
accepts_online_orders	boolean	No	false	-	-	Merchant online-order toggle
service_radius_meters	integer	No	5000	CHECK	-	Delivery service radius
minimum_order_paise	bigint	No	0	CHECK	-	Minimum order value
average_preparation_m_inutes	integer	No	15	CHECK	-	Estimated preparation time
rating_average	numeric(3, 2)	No	0	CHECK	-	Cached average
rating_count	integer	No	0	CHECK	-	Cached count
follower_count	integer	No	0	CHECK	-	Cached favourites/followers
version	integer	No	1	CHECK	-	Optimistic locking version
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp
deleted_at	timestamptz	Yes	-	-	-	Soft delete

shop_hours [MVP]

Weekly and date-specific opening hours.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Schedule row
shop_id	uuid	No	-	FK	shops.id	Shop
schedule_type	text	No	'WEEKLY'	CHECK	-	WEEKLY or SPECIAL_DATE
day_of_week	smallint	Yes	-	CHECK	-	0-6 for weekly schedule
special_date	date	Yes	-	-	-	Date override
open_time	time	Yes	-	-	-	Local opening time
close_time	time	Yes	-	-	-	Local closing time
is_closed	boolean	No	false	-	-	Closed flag
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

Additional constraints: unique (shop_id, schedule_type, day_of_week, special_date)

shop_documents [MVP]

Private merchant/shop KYC files and verification status.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Document ID
shop_id	uuid	No	-	FK	shops.id	Shop
uploaded_by	uuid	No	-	FK	profiles.id	Merchant or field executive
document_type	text	No	-	CHECK	-	PAN, GST, LICENSE, ADDRESS_PROOF, OWNER_ID, SHOP_PHOTO, OTHER
document_number_last4	text	Yes	-	-	-	Masked display value
document_number_encrypted	text	Yes	-	-	-	Encrypted sensitive number
storage_path	text	No	-	-	-	Private Storage object path
verification_status	text	No	'PENDING'	CHECK	-	PENDING, VERIFIED, REJECTED
verified_by	uuid	Yes	-	FK	profiles.id	Reviewing admin
verified_at	timestamptz	Yes	-	-	-	Verification timestamp
rejection_reason	text	Yes	-	-	-	Correction or rejection reason
created_at	timestamptz	No	now()	-	-	Creation timestamp

shop_bank_accounts [MVP]

Private merchant payout account records.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Bank account ID
shop_id	uuid	No	-	FK	shops.id	Shop
account_holder_name	text	No	-	-	-	Account holder
account_number_last4	text	No	-	-	-	Masked display digits

Attribute	Type	Null?	Default	Key	Reference	Purpose
account_number_encrypted	text	No	-	-	-	Encrypted account number
ifsc_code	text	No	-	-	-	Bank IFSC
bank_name	text	Yes	-	-	-	Bank name
branch_name	text	Yes	-	-	-	Branch
is_primary	boolean	No	true	-	-	Primary payout destination
verification_status	text	No	'PENDING'	CHECK	-	PENDING, VERIFIED, REJECTED
verified_at	timestampz	Yes	-	-	-	Verification timestamp
created_at	timestampz	No	now()	-	-	Creation timestamp
updated_at	timestampz	No	now()	-	-	Update timestamp

Catalogue

7 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

categories [MVP]

Hierarchical catalogue taxonomy covering clothing, footwear and accessories.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Category ID
parent_id	uuid	Yes	-	FK	categories.id	Parent category
name	text	No	-	-	-	Category display name
slug	text	No	-	UNIQUE	-	URL/search identifier
description	text	Yes	-	-	-	Category description
icon_path	text	Yes	-	-	-	Icon object path
display_order	integer	No	0	-	-	UI sort order
is_active	boolean	No	true	-	-	Visibility flag
created_at	timestampz	No	now()	-	-	Creation timestamp
updated_at	timestampz	No	now()	-	-	Update timestamp

products [MVP]

General merchant product record independent of colour/size variants.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Product ID
shop_id	uuid	No	-	FK	shops.id	Selling shop
category_id	uuid	No	-	FK	categories.id	Leaf category
name	text	No	-	-	-	Product name
slug	text	No	-	-	-	Shop-scoped URL identifier
description	text	Yes	-	-	-	Detailed description

Attribute	Type	Null?	Default	Key	Reference	Purpose
brand	text	Yes	-	-	-	Brand/local label
material	text	Yes	-	-	-	Cotton, silk, leather, etc.
gender_category	text	No	'UNISEX'	CHECK	-	MEN, WOMEN, KIDS, UNISEX
style_tags	text[]	No	'{'	-	-	Casual, formal, traditional, etc.
occasion_tags	text[]	No	'{'	-	-	Wedding, party, trip, festival, etc.
care_instructions	text	Yes	-	-	-	Washing/care instructions
return_eligible	boolean	No	true	-	-	Return eligibility
return_window_days	smallint	No	7	CHECK	-	Return window
moderation_status	text	No	'PENDING'	CHECK	-	PENDING, APPROVED, REJECTED, CORRECTION_REQUIRED
is_active	boolean	No	true	-	-	Merchant availability flag
search_vector	tsvector	Yes	-	-	-	Generated/full-text search document
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp
deleted_at	timestamptz	Yes	-	-	-	Soft delete

Additional constraints: unique (shop_id, slug)

product_images [MVP]

Ordered product and variant media references stored in Supabase Storage.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Image ID
product_id	uuid	No	-	FK	products.id	Product
variant_id	uuid	Yes	-	FK	product_variants.id	Variant-specific image when applicable
storage_path	text	No	-	-	-	Object path
thumbnail_path	text	Yes	-	-	-	Optimized thumbnail path
image_type	text	No	'FRONT'	CHECK	-	FRONT, BACK, SIDE, DETAIL, MODEL, SIZE_CHART
alt_text	text	Yes	-	-	-	Accessibility text
display_order	smallint	No	0	-	-	Sort order
is_primary	boolean	No	false	-	-	Primary image flag
width_px	integer	Yes	-	-	-	Source width
height_px	integer	Yes	-	-	-	Source height
created_at	timestamptz	No	now()	-	-	Creation timestamp

product_variants [MVP]

Exact sellable SKU for each colour, size and optional attribute combination.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Variant ID
product_id	uuid	No	-	FK	products.id	Parent product
shop_id	uuid	No	-	FK	shops.id	Denormalized shop for filtering/RLS

Attribute	Type	Null?	Default	Key	Reference	Purpose
sku	text	No	-	UNIQUE-PART	-	Merchant/Vastra SKU
colour_name	text	Yes	-	-	-	Colour label
colour_hex	text	Yes	-	-	-	UI colour swatch
size_label	text	Yes	-	-	-	S, M, L, shoe 9, belt 34, etc.
mrp_paise	bigint	No	-	CHECK	-	Maximum retail price
selling_price_paise	bigint	No	-	CHECK	-	Current selling price
cost_price_paise	bigint	Yes	-	CHECK	-	Optional private cost
weight_grams	integer	Yes	-	CHECK	-	Logistics weight
length_cm	numeric(8,2)	Yes	-	CHECK	-	Package/garment length
width_cm	numeric(8,2)	Yes	-	CHECK	-	Width
height_cm	numeric(8,2)	Yes	-	CHECK	-	Height
attributes	jsonb	No	'{}'::jsonb	-	-	Flexible category-specific attributes
is_active	boolean	No	true	-	-	Variant sellable flag
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

Additional constraints: unique (shop_id, sku); check (selling_price_paise <= mrp_paise)

variant_barcodes [MVP]

Manufacturer, merchant-entered and Vastra-generated barcodes mapped to exact variants.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Barcode mapping ID
variant_id	uuid	No	-	FK	product_variants.id	Exact variant
barcode_value	text	No	-	UNIQUE	-	Encoded value
barcode_type	text	No	'CODE128'	CHECK	-	EAN13, UPC, CODE128, QR or INTERNAL
source	text	No	'MERCHANT_ENTERED'	CHECK	-	MANUFACTURER, VASTRA_GENERATED, MERCHANT_ENTERED
is_primary	boolean	No	true	-	-	Preferred scan/print code
created_by	uuid	Yes	-	FK	profiles.id	Creator
created_at	timestamptz	No	now()	-	-	Creation timestamp

shop_collections [MVP]

Merchant-curated occasion, complete-look and promotional collections.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Collection ID
shop_id	uuid	No	-	FK	shops.id	Shop
name	text	No	-	-	-	Collection name
description	text	Yes	-	-	-	Collection description
collection_type	text	No	'MANUAL'	CHECK	-	MANUAL, COMPLETE_LOOK, OCCASION, GROUP_STYLE

Attribute	Type	Null?	Default	Key	Reference	Purpose
cover_image_path	text	Yes	-	-	-	Cover image
is_active	boolean	No	true	-	-	Visibility
display_order	integer	No	0	-	-	Shop page order
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

shop_collection_items [MVP]

Ordered products/variants inside merchant collections.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Item ID
collection_id	uuid	No	-	FK	shop_collections.id	Collection
product_id	uuid	No	-	FK	products.id	Product
variant_id	uuid	Yes	-	FK	product_variants.id	Optional exact variant
item_role	text	Yes	-	-	-	MAIN, BOTTOM, FOOTWEAR, BELT, WATCH, BAG, ACCESSORY
display_order	integer	No	0	-	-	Sort order
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (collection_id, product_id, variant_id)

Inventory

4 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

inventory_balances [MVP]

Current variant-level physical, reserved and damaged quantities. This is the authoritative balance row.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Balance ID
shop_id	uuid	No	-	FK	shops.id	Shop
variant_id	uuid	No	-	FK	product_variants.id	Variant
stock_on_hand	integer	No	0	CHECK	-	Physical units including reserved units
reserved_quantity	integer	No	0	CHECK	-	Units reserved for active online orders
damaged_quantity	integer	No	0	CHECK	-	Unsellable damaged units
reorder_level	integer	No	0	CHECK	-	Low-stock threshold
version	integer	No	1	CHECK	-	Optimistic locking version
last_counted_at	timestamptz	Yes	-	-	-	Last physical audit
updated_at	timestamptz	No	now()	-	-	Update timestamp

Additional constraints: unique (shop_id, variant_id); check (stock_on_hand >= reserved_quantity + damaged_quantity)

inventory_movements [MVP]

Immutable audit ledger for every stock change from barcode, photo, manual action, order or admin correction.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Movement sequence
shop_id	uuid	No	-	FK	shops.id	Shop
variant_id	uuid	No	-	FK	product_variants.id	Variant
movement_type	text	No	-	CHECK	-	STOCK_RECEIVED, OFFLINE_SALE, ONLINE_ORDER_RESERVED, ONLINE_ORDER_RELEASED, ONLINE_ORDER_COMPLETED, RETURN_TO_STOCK, MARKED_DAMAGED, STOCK_CORRECTION, STOCK_AUDIT
quantity_change	integer	No	-	-	-	Signed physical quantity change
reserved_change	integer	No	0	-	-	Signed reservation change
damaged_change	integer	No	0	-	-	Signed damaged quantity change
stock_before	integer	No	-	-	-	Before snapshot
stock_after	integer	No	-	-	-	After snapshot
reserved_before	integer	No	-	-	-	Before reservation
reserved_after	integer	No	-	-	-	After reservation
damaged_before	integer	No	-	-	-	Before damaged quantity
damaged_after	integer	No	-	-	-	After damaged quantity
reference_type	text	Yes	-	-	-	ORDER, OFFLINE_SALE, RETURN, AUDIT, MANUAL
reference_id	uuid	Yes	-	-	-	Related business entity
reason	text	Yes	-	-	-	Human/audit explanation
performed_by	uuid	Yes	-	FK	profiles.id	Actor
source_method	text	No	'SYSTEM'	CHECK	-	BARCODE, PHOTO_MATCH, MANUAL_SEARCH, SYSTEM, ADMIN
created_at	timestamptz	No	now()	-	-	Immutable timestamp

inventory_reservations [MVP]

Temporary or order-linked reservations that prevent double-selling the final unit.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Reservation ID
shop_id	uuid	No	-	FK	shops.id	Shop
variant_id	uuid	No	-	FK	product_variants.id	Variant
cart_id	uuid	Yes	-	FK	carts.id	Checkout reservation when used
order_id	uuid	Yes	-	FK	orders.id	Order reservation after placement
quantity	integer	No	-	CHECK	-	Reserved units
status	text	No	'ACTIVE'	CHECK	-	ACTIVE, CONVERTED, RELEASED, EXPIRED
expires_at	timestamptz	No	-	-	-	Reservation expiry
created_at	timestamptz	No	now()	-	-	Creation timestamp
released_at	timestamptz	Yes	-	-	-	Release timestamp

product_photo_matches [Phase 2]

History and feedback for merchant photo-based catalogue matching.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Match session ID
shop_id	uuid	No	-	FK	shops.id	Merchant catalogue searched
uploaded_by	uuid	No	-	FK	profiles.id	Merchant
uploaded_image_path	text	No	-	-	-	Temporary/private image path
matched_variant_id	uuid	Yes	-	FK	product_variants.id	Merchant-selected result
confidence_score	numeric(6,5)	Yes	-	CHECK	-	Model confidence
candidate_matches	jsonb	No	'[]'::jsonb	-	-	Ranked candidates and scores
merchant_confirmed	boolean	No	false	-	-	Ground-truth feedback
match_model_version	text	Yes	-	-	-	Model/version identifier
created_at	timestamptz	No	now()	-	-	Creation timestamp

Commerce

10 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

offline_sales [MVP]

Physical shop sale recorded by barcode, photo match or manual product selection so online inventory stays accurate.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Offline sale ID
sale_number	text	No	-	UNIQUE	-	Merchant-facing sale number
shop_id	uuid	No	-	FK	shops.id	Shop
merchant_id	uuid	No	-	FK	merchant_profiles.user_id	Merchant account
customer_phone	text	Yes	-	-	-	Optional contact, collected only with consent
subtotal_paise	bigint	No	0	CHECK	-	Item subtotal
discount_paise	bigint	No	0	CHECK	-	Discount
tax_paise	bigint	No	0	CHECK	-	Tax if applicable
total_paise	bigint	No	0	CHECK	-	Final amount
payment_method	text	No	-	CHECK	-	CASH, UPI, CARD, OTHER
status	text	No	'COMPLETED'	CHECK	-	DRAFT, COMPLETED, VOIDED, REFUNDED
recorded_by	uuid	No	-	FK	profiles.id	Merchant actor
created_at	timestamptz	No	now()	-	-	Sale timestamp
voided_at	timestamptz	Yes	-	-	-	Void timestamp

offline_sale_items [MVP]

Line items in a physical shop sale.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Line ID
offline_sale_id	uuid	No	-	FK	offline_sales.id	Sale
variant_id	uuid	No	-	FK	product_variants.id	Exact item
quantity	integer	No	-	CHECK	-	Units
unit_price_paise	bigint	No	-	CHECK	-	Selling price snapshot
discount_paise	bigint	No	0	CHECK	-	Line discount
total_paise	bigint	No	-	CHECK	-	Line total
identification_method	text	No	'MANUAL_SEARCH'	CHECK	-	BARCODE, PHOTO_MATCH, MANUAL_SEARCH
created_at	timestamptz	No	now()	-	-	Creation timestamp

carts [MVP]

One active single-shop cart per customer in the MVP.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Cart ID
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
shop_id	uuid	No	-	FK	shops.id	Merchant/shop
status	text	No	'ACTIVE'	CHECK	-	ACTIVE, CONVERTED, ABANDONED, EXPIRED
coupon_code	text	Yes	-	-	-	Applied coupon code
expires_at	timestamptz	Yes	-	-	-	Optional cart expiry
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

cart_items [MVP]

Variant-level cart lines with price snapshots for display; final prices are revalidated at order placement.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Cart line ID
cart_id	uuid	No	-	FK	carts.id	Cart
variant_id	uuid	No	-	FK	product_variants.id	Variant
quantity	integer	No	1	CHECK	-	Requested units
unit_price_snapshot_paise	bigint	No	-	CHECK	-	Display price snapshot
added_at	timestamptz	No	now()	-	-	Added timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

Additional constraints: unique (cart_id, variant_id)

orders [MVP]

Authoritative customer order header and current lifecycle state.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Order ID
order_number	text	No	-	UNIQUE	-	Customer-facing order number
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
shop_id	uuid	No	-	FK	shops.id	Shop
cart_id	uuid	Yes	-	FK	carts.id	Source cart
style_group_id	uuid	Yes	-	FK	style_groups.id	Optional Group Style source
delivery_address_id	uuid	No	-	FK	addresses.id	Chosen customer address
address_snapshot	jsonb	No	'{}'::jsonb	-	-	Immutable delivery address snapshot
status	text	No	'PAYMENT_PENDING'	CHECK	-	Current order state
payment_status	text	No	'PENDING'	CHECK	-	PENDING, AUTHORIZED, CAPTURED, FAILED, PARTIALLY_REFUNDED, REFUNDED, COD_PENDING, COD_COLLECTED
fulfilment_type	text	No	'DELIVERY'	CHECK	-	DELIVERY or CUSTOMER_PICKUP
subtotal_paise	bigint	No	0	CHECK	-	Item subtotal
product_discount_paise	bigint	No	0	CHECK	-	Product discount
coupon_discount_paise	bigint	No	0	CHECK	-	Coupon discount
delivery_fee_paise	bigint	No	0	CHECK	-	Delivery fee
platform_fee_paise	bigint	No	0	CHECK	-	Platform fee
tax_paise	bigint	No	0	CHECK	-	Tax
total_paise	bigint	No	0	CHECK	-	Final payable
merchant_preparation_minutes	integer	Yes	-	CHECK	-	Accepted preparation time
estimated_delivery_at	timestamptz	Yes	-	-	-	Current ETA
customer_note	text	Yes	-	-	-	Checkout note
cancellation_reason_code	text	Yes	-	-	-	Structured reason
cancellation_note	text	Yes	-	-	-	Additional reason
version	integer	No	1	CHECK	-	Optimistic locking version
placed_at	timestamptz	Yes	-	-	-	Placement timestamp
accepted_at	timestamptz	Yes	-	-	-	Merchant acceptance
ready_at	timestamptz	Yes	-	-	-	Ready-for-pickup timestamp
picked_up_at	timestamptz	Yes	-	-	-	Pickup timestamp
delivered_at	timestamptz	Yes	-	-	-	Delivery timestamp
cancelled_at	timestamptz	Yes	-	-	-	Cancellation timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

order_items [MVP]

Immutable product and price snapshots for each exact ordered variant.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Order item ID
order_id	uuid	No	-	FK	orders.id	Order
product_id	uuid	No	-	FK	products.id	Product
variant_id	uuid	No	-	FK	product_variants.id	Variant
product_name_snapshot	text	No	-	-	-	Product name at order time
sku_snapshot	text	No	-	-	-	SKU at order time
colour_snapshot	text	Yes	-	-	-	Colour
size_snapshot	text	Yes	-	-	-	Size
image_path_snapshot	text	Yes	-	-	-	Image path
quantity	integer	No	-	CHECK	-	Units
unit_mrp_paise	bigint	No	-	CHECK	-	MRP snapshot
unit_selling_price_paise	bigint	No	-	CHECK	-	Selling price snapshot
discount_paise	bigint	No	0	CHECK	-	Line discount
total_paise	bigint	No	-	CHECK	-	Line total
fulfilment_status	text	No	'PENDING'	CHECK	-	PENDING, VERIFIED, PACKED, HANDED_OVER, RETURNED, CANCELLED
created_at	timestamptz	No	now()	-	-	Creation timestamp

order_status_history [MVP]

Append-only timeline for every state transition and operational reason.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Timeline sequence
order_id	uuid	No	-	FK	orders.id	Order
previous_status	text	Yes	-	-	-	Previous state
new_status	text	No	-	-	-	New state
changed_by_user_id	uuid	Yes	-	FK	profiles.id	Actor or null for system
changed_by_role	text	No	'SYSTEM'	-	-	CUSTOMER, MERCHANT, CAPTAIN, ADMIN, SYSTEM
reason_code	text	Yes	-	-	-	Structured reason
note	text	Yes	-	-	-	Internal/customer-visible note
location	geography(Point, 4326)	Yes	-	-	-	Location when operationally relevant
created_at	timestamptz	No	now()	-	-	Transition timestamp

merchant_order_alerts [MVP]

Tracks every urgent order alert, retry, delivery and acknowledgement for the merchant ringing experience.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Alert ID
order_id	uuid	No	-	FK	orders.id	New order
shop_id	uuid	No	-	FK	shops.id	Target shop
device_id	uuid	Yes	-	FK	user_devices.id	Target device when known
alert_status	text	No	'PENDING'	CHECK	-	PENDING, SENT, DELIVERED, ACKNOWLEDGED, EXPIRED, FAILED
attempt_count	smallint	No	0	CHECK	-	Number of push/realtime attempts
first_sent_at	timestamptz	Yes	-	-	-	First send
last_sent_at	timestamptz	Yes	-	-	-	Last retry
acknowledged_at	timestamptz	Yes	-	-	-	Merchant acknowledgement
acknowledged_by	uuid	Yes	-	FK	profiles.id	Merchant
expires_at	timestamptz	No	-	-	-	Response deadline
sound_name	text	No	'vastra_new_order'	-	-	Notification channel/sound identifier
failure_reason	text	Yes	-	-	-	Push/provider failure detail
created_at	timestamptz	No	now()	-	-	Creation timestamp

merchant_order_issues [MVP]

Merchant-reported inability to fulfil an order because of stock, size, colour, damage, pricing or operational issues.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Issue ID
order_id	uuid	No	-	FK	orders.id	Order
order_item_id	uuid	Yes	-	FK	order_items.id	Affected item
issue_type	text	No	-	CHECK	-	OUT_OF_STOCK, SIZE_UNAVAILABLE, COLOUR_UNAVAILABLE, DAMAGED_ITEM, INVENTORY_MISMATCH, ITEM_NOT_FOUND, PRICING_ERROR, SHOP_BUSY, SHOP_CLOSING, OTHER
description	text	Yes	-	-	-	Merchant note
evidence_path	text	Yes	-	-	-	Private Storage evidence
reported_by	uuid	No	-	FK	profiles.id	Merchant
resolution_status	text	No	'OPEN'	CHECK	-	OPEN, ACCEPTED, REPLACEMENT_REQUESTED, RESOLVED, REJECTED
resolved_by	uuid	Yes	-	FK	profiles.id	Admin/system resolver
resolution_note	text	Yes	-	-	-	Resolution
created_at	timestamptz	No	now()	-	-	Report timestamp
resolved_at	timestamptz	Yes	-	-	-	Resolution timestamp

order_item_verifications [MVP]

Packing verification result using barcode, photo or manual confirmation.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Verification ID
order_item_id	uuid	No	-	FK	order_items.id	Item
verification_method	text	No	-	CHECK	-	BARCODE, PHOTO, MANUAL
scanned_barcode	text	Yes	-	-	-	Barcode read
photo_path	text	Yes	-	-	-	Temporary/private image
verified_variant_id	uuid	Yes	-	FK	product_variants.id	Detected/selected variant
result	text	No	-	CHECK	-	MATCH, MISMATCH, OVERRIDDEN
verified_by	uuid	No	-	FK	profiles.id	Merchant
verified_at	timestamptz	No	now()	-	-	Verification timestamp
override_reason	text	Yes	-	-	-	Required for manual override

Payments

3 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

payments [MVP]

Payment attempt and provider identifiers for online and COD orders.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Payment ID
order_id	uuid	No	-	FK	orders.id	Order
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
provider	text	No	-	-	-	CASHFREE, RAZORPAY, COD, etc.
provider_order_id	text	Yes	-	-	-	Gateway order/reference ID
provider_payment_id	text	Yes	-	-	-	Gateway payment ID
method	text	No	-	CHECK	-	UPI, CARD, NETBANKING, WALLET, COD, OTHER
amount_paise	bigint	No	-	CHECK	-	Amount
currency	text	No	'INR'	-	-	Currency
status	text	No	'CREATED'	CHECK	-	CREATED, PENDING, AUTHORIZED, CAPTURED, FAILED, CANCELLED, PARTIALLY_REFUNDED, REFUNDED
signature_verified	boolean	No	false	-	-	Server-side verification status
failure_code	text	Yes	-	-	-	Provider failure code
failure_message	text	Yes	-	-	-	Provider failure message
paid_at	timestamptz	Yes	-	-	-	Captured timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

payment_events [MVP]

Idempotent raw gateway webhook/event log.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Event sequence
payment_id	uuid	Yes	-	FK	payments.id	Payment when resolved
provider	text	No	-	-	-	Gateway
provider_event_id	text	No	-	UNIQUE	-	Idempotency key from provider
event_type	text	No	-	-	-	Provider event type
payload	jsonb	No	-	-	-	Original payload after signature verification
signature_valid	boolean	No	false	-	-	Verification outcome
processing_status	text	No	'RECEIVED'	CHECK	-	RECEIVED, PROCESSED, IGNORED, FAILED
processing_error	text	Yes	-	-	-	Error detail
received_at	timestamptz	No	now()	-	-	Receipt timestamp
processed_at	timestamptz	Yes	-	-	-	Processing timestamp

refunds [MVP]

Gateway/manual refund record linked to order, payment and optional return.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Refund ID
refund_number	text	No	-	UNIQUE	-	Customer/admin-facing number
order_id	uuid	No	-	FK	orders.id	Order
payment_id	uuid	Yes	-	FK	payments.id	Original payment
return_request_id	uuid	Yes	-	FK	return_requests.id	Return when applicable
amount_paise	bigint	No	-	CHECK	-	Refund amount
reason_code	text	No	-	-	-	Reason
provider_refund_id	text	Yes	-	-	-	Gateway refund ID
status	text	No	'PENDING'	CHECK	-	PENDING, APPROVAL_REQUIRED, INITIATED, PROCESSING, COMPLETED, FAILED, CANCELLED
initiated_by	uuid	No	-	FK	profiles.id	Requesting actor
approved_by	uuid	Yes	-	FK	profiles.id	Approver
initiated_at	timestamptz	Yes	-	-	-	Initiation timestamp
completed_at	timestamptz	Yes	-	-	-	Completion timestamp
failure_message	text	Yes	-	-	-	Failure detail
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

Returns

4 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

return_requests [MVP]

Customer return request header and current workflow status.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Return ID
return_number	text	No	-	UNIQUE	-	Customer-facing return number
order_id	uuid	No	-	FK	orders.id	Original order
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
shop_id	uuid	No	-	FK	shops.id	Merchant
reason_code	text	No	-	CHECK	-	WRONG_SIZE, WRONG_COLOUR, WRONG_PRODUCT, DAMAGED, QUALITY, DIFFERENT_FROM_IMAGE, MISSING_ITEM, CHANGED_MIND, OTHER
customer_note	text	Yes	-	-	-	Customer explanation
status	text	No	'REQUESTED'	CHECK	-	REQUESTED, REVIEW, APPROVED, REJECTED, PICKUP_ASSIGNED, PICKED_UP, RECEIVED, VERIFIED, REFUND_PENDING, REFUNDED, CLOSED
refund_method	text	No	'ORIGINAL'	CHECK	-	ORIGINAL, WALLET, MANUAL
requested_at	timestamptz	No	now()	-	-	Request timestamp
approved_at	timestamptz	Yes	-	-	-	Approval timestamp
completed_at	timestamptz	Yes	-	-	-	Completion timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

return_items [MVP]

Per-order-item quantities, inspection and merchant decision.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Return item ID
return_request_id	uuid	No	-	FK	return_requests.id	Return
order_item_id	uuid	No	-	FK	order_items.id	Original item
quantity	integer	No	1	CHECK	-	Quantity
reason_code	text	No	-	-	-	Item reason
requested_refund_paise	bigint	No	-	CHECK	-	Requested amount
inspection_status	text	No	'PENDING'	CHECK	-	PENDING, SELLABLE, DAMAGED, USED, WRONG_ITEM, DISPUTED
merchant_decision	text	Yes	-	CHECK	-	ACCEPTED, DISPUTED, PARTIAL, null until reviewed
merchant_note	text	Yes	-	-	-	Merchant inspection note
inspected_by	uuid	Yes	-	FK	profiles.id	Merchant/admin
inspected_at	timestamptz	Yes	-	-	-	Inspection timestamp

return_evidence [MVP]

Customer, merchant, captain and admin evidence files for return disputes.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Evidence ID
return_request_id	uuid	No	-	FK	return_requests.id	Return
uploaded_by	uuid	No	-	FK	profiles.id	Actor
evidence_type	text	No	-	CHECK	-	CUSTOMER_PHOTO, MERCHANT_PHOTO, CAPTAIN_PHOTO, VIDEO, DOCUMENT, NOTE
storage_path	text	Yes	-	-	-	Private file path
description	text	Yes	-	-	-	Evidence description
created_at	timestampz	No	now()	-	-	Creation timestamp

return_status_history [MVP]

Append-only return state timeline.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Timeline sequence
return_request_id	uuid	No	-	FK	return_requests.id	Return
previous_status	text	Yes	-	-	-	Previous state
new_status	text	No	-	-	-	New state
changed_by	uuid	Yes	-	FK	profiles.id	Actor
reason_code	text	Yes	-	-	-	Reason
note	text	Yes	-	-	-	Note
created_at	timestampz	No	now()	-	-	Timestamp

Finance

4 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

merchant_settlements [MVP]

Periodic merchant payout batch with commissions, discount share, refunds and adjustments.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Settlement ID
settlement_number	text	No	-	UNIQUE	-	Settlement number
shop_id	uuid	No	-	FK	shops.id	Shop
bank_account_id	uuid	No	-	FK	shop_bank_accounts.id	Payout destination
period_start	date	No	-	-	-	Settlement period
period_end	date	No	-	-	-	Settlement period
gross_sales_paise	bigint	No	0	-	-	Gross delivered sales

Attribute	Type	Null?	Default	Key	Reference	Purpose
commission_paise	bigint	No	0	-	-	Platform commission
discount_share_paise	bigint	No	0	-	-	Merchant-funded promotions
refund_deduction_paise	bigint	No	0	-	-	Refund deductions
tax_paise	bigint	No	0	-	-	Applicable tax withholding
adjustment_paise	bigint	No	0	-	-	Signed adjustments
net_payable_paise	bigint	No	0	-	-	Final payout
status	text	No	'DRAFT'	CHECK	-	DRAFT, REVIEW, APPROVED, PROCESSING, PAID, FAILED, ON_HOLD
provider_transfer_id	text	Yes	-	-	-	Payout provider ID
scheduled_at	timestamptz	Yes	-	-	-	Scheduled payout
paid_at	timestamptz	Yes	-	-	-	Payment timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

merchant_settlement_items [MVP]

Settlement ledger rows linking orders, refunds and manual adjustments.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Settlement item ID
settlement_id	uuid	No	-	FK	merchant_settlements.id	Settlement
order_id	uuid	Yes	-	FK	orders.id	Related order
refund_id	uuid	Yes	-	FK	refunds.id	Related refund
entry_type	text	No	-	CHECK	-	ORDER_CREDIT, COMMISSION, DISCOUNT, REFUND, TAX, PENALTY, MANUAL_ADJUSTMENT
amount_paise	bigint	No	-	-	-	Signed amount
description	text	Yes	-	-	-	Explanation
created_at	timestamptz	No	now()	-	-	Creation timestamp

captain_payouts [MVP]

Periodic captain payout batch including earnings, incentives, penalties and COD adjustments.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Payout ID
payout_number	text	No	-	UNIQUE	-	Payout number
captain_id	uuid	No	-	FK	captain_profiles.user_id	Captain
period_start	date	No	-	-	-	Period
period_end	date	No	-	-	-	Period
earnings_paise	bigint	No	0	-	-	Delivery earnings
incentives_paise	bigint	No	0	-	-	Incentives
penalties_paise	bigint	No	0	-	-	Penalties

Attribute	Type	Null?	Default	Key	Reference	Purpose
cod_adjustment_paise	bigint	No	0	-	-	COD cash offset
net_payout_paise	bigint	No	0	-	-	Net payout
status	text	No	'DRAFT'	CHECK	-	DRAFT, REVIEW, APPROVED, PROCESSING, PAID, FAILED, ON_HOLD
provider_transfer_id	text	Yes	-	-	-	Transfer identifier
paid_at	timestamptz	Yes	-	-	-	Paid timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

captain_payout_items [MVP]

Links a payout batch to individual captain earning records and adjustments.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Payout item ID
payout_id	uuid	No	-	FK	captain_payouts.id	Payout
captain_earning_id	uuid	Yes	-	FK	captain_earnings.id	Earning
entry_type	text	No	-	CHECK	-	EARNING, INCENTIVE, PENALTY, COD_ADJUSTMENT, MANUAL_ADJUSTMENT
amount_paise	bigint	No	-	-	-	Signed amount
description	text	Yes	-	-	-	Explanation
created_at	timestamptz	No	now()	-	-	Creation timestamp

Logistics

8 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

captain_documents [MVP]

Private KYC, licence, vehicle and bank documents for captains.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Document ID
captain_id	uuid	No	-	FK	captain_profiles.user_id	Captain
document_type	text	No	-	CHECK	-	AADHAAR, PAN, DRIVING_LICENCE, VEHICLE_RC, INSURANCE, BANK_PROOF, PROFILE_PHOTO, OTHER
storage_path	text	No	-	-	-	Private Storage path
document_number_last4	text	Yes	-	-	-	Masked display
document_number_encrypted	text	Yes	-	-	-	Encrypted number
expiry_date	date	Yes	-	-	-	Expiry when applicable
verification_status	text	No	'PENDING'	CHECK	-	PENDING, VERIFIED, REJECTED, EXPIRED
verified_by	uuid	Yes	-	FK	profiles.id	Admin reviewer
verified_at	timestamptz	Yes	-	-	-	Verification timestamp
rejection_reason	text	Yes	-	-	-	Reason

Attribute	Type	Null?	Default	Key	Reference	Purpose
created_at	timestamptz	No	now()	-	-	Creation timestamp

delivery_tasks [MVP]

Reusable delivery job for forward delivery, return pickup, return-to-merchant and later exchange flows.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Delivery task ID
order_id	uuid	Yes	-	FK	orders.id	Forward order
return_request_id	uuid	Yes	-	FK	return_requests.id	Return workflow
task_type	text	No	'FORWARD_DELIVERY'	CHECK	-	FORWARD_DELIVERY, RETURN_PICKUP, RETURN_TO_MERCHANT, EXCHANGE_DELIVERY
pickup_shop_id	uuid	Yes	-	FK	shops.id	Pickup shop
pickup_address_snapshot	jsonb	No	'{}'::jsonb	-	-	Immutable pickup details
drop_address_snapshot	jsonb	No	'{}'::jsonb	-	-	Immutable drop details
pickup_location	geography(Point, 4326)	No	-	-	-	Pickup point
drop_location	geography(Point, 4326)	No	-	-	-	Drop point
status	text	No	'CREATED'	CHECK	-	CREATED, SEARCHING, OFFERED, ASSIGNED, AT_PICKUP, PICKED_UP, IN_TRANSIT, AT_DROP, COMPLETED, FAILED, CANCELLED
estimated_distance_meters	integer	Yes	-	CHECK	-	Estimated route distance
estimated_duration_seconds	integer	Yes	-	CHECK	-	Estimated duration
delivery_fee_paise	bigint	No	0	CHECK	-	Customer/merchant delivery fee
captain_earning_paise	bigint	No	0	CHECK	-	Base expected captain earning
pickup_code_hash	text	Yes	-	-	-	Never store plain pickup code
delivery_otp_hash	text	Yes	-	-	-	Never store plain delivery OTP
assigned_captain_id	uuid	Yes	-	FK	captain_profiles.user_id	Current captain
assignment_attempts	smallint	No	0	CHECK	-	Dispatch attempts
scheduled_at	timestamptz	Yes	-	-	-	Scheduled delivery/return
assigned_at	timestamptz	Yes	-	-	-	Assignment timestamp
picked_up_at	timestamptz	Yes	-	-	-	Pickup timestamp
completed_at	timestamptz	Yes	-	-	-	Completion timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

delivery_assignments [MVP]

Complete dispatch offer history including rejections, timeouts and reassignments.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Assignment ID

Attribute	Type	Null?	Default	Key	Reference	Purpose
delivery_task_id	uuid	No	-	FK	delivery_tasks.id	Task
captain_id	uuid	No	-	FK	captain_profiles.user_id	Offered captain
assignment_status	text	No	'OFFERED'	CHECK	-	OFFERED, ACCEPTED, REJECTED, TIMED_OUT, CANCELLED, RELEASED, COMPLETED
offered_earning_paise	bigint	No	0	CHECK	-	Amount shown to captain
pickup_distance_meters	integer	Yes	-	CHECK	-	Captain-to-pickup distance
offered_at	timestamptz	No	now()	-	-	Offer timestamp
expires_at	timestamptz	No	-	-	-	Response deadline
responded_at	timestamptz	Yes	-	-	-	Response time
rejection_reason	text	Yes	-	-	-	Captain reason
assigned_by	text	No	'AUTO'	CHECK	-	AUTO or ADMIN
assigned_by_user_id	uuid	Yes	-	FK	profiles.id	Admin manual assignment
created_at	timestamptz	No	now()	-	-	Creation timestamp

captain_current_locations [MVP]

Latest captain location for dispatch and customer tracking; hot copies may also live in Redis.

Attribute	Type	Null?	Default	Key	Reference	Purpose
captain_id	uuid	No	-	PK/FK	captain_profiles.user_id	Captain
location	geography(Point, 4326)	No	-	-	-	Latest point
heading	numeric(6, 2)	Yes	-	-	-	Direction degrees
speed_mps	numeric(8, 2)	Yes	-	CHECK	-	Speed
accuracy_meters	numeric(8, 2)	Yes	-	CHECK	-	Device accuracy
battery_percent	smallint	Yes	-	CHECK	-	Optional device battery
active_delivery_task_id	uuid	Yes	-	FK	delivery_tasks.id	Active task
updated_at	timestamptz	No	now()	-	-	Location timestamp

captain_location_history [MVP]

Durable sampled location trail for active deliveries and dispute investigation. Partition monthly.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Location sequence
captain_id	uuid	No	-	FK	captain_profiles.user_id	Captain
delivery_task_id	uuid	Yes	-	FK	delivery_tasks.id	Active task
location	geography(Point, 4326)	No	-	-	-	Sampled point
heading	numeric(6, 2)	Yes	-	-	-	Direction
speed_mps	numeric(8, 2)	Yes	-	-	-	Speed

Attribute	Type	Null?	Default	Key	Reference	Purpose
accuracy_meters	numeric(8,2)	Yes	-	-	-	Accuracy
recorded_at	timestamptz	No	now()	-	-	Device timestamp
received_at	timestamptz	No	now()	-	-	Server timestamp

delivery_events [MVP]

Append-only operational event log for pickup, delays, failures, arrival and completion.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Event sequence
delivery_task_id	uuid	No	-	FK	delivery_tasks.id	Task
event_type	text	No	-	CHECK	-	CAPTAIN_ASSIGNED, ARRIVED_AT_STORE, MERCHANT_DELAY, PICKUP_CONFIRMED, LEFT_STORE, ARRIVED_AT_CUSTOMER, DELIVERY_CONFIRMED, CUSTOMER_UNAVAILABLE, INVALID_ADDRESS, CUSTOMER_REFUSED, PACKAGE_DAMAGED, RETURNING_TO_STORE, TASK_COMPLETED, OTHER
actor_user_id	uuid	Yes	-	FK	profiles.id	Captain/admin/system actor
location	geography(Point,4326)	Yes	-	-	-	Event location
note	text	Yes	-	-	-	Event detail
evidence_path	text	Yes	-	-	-	Private evidence
metadata	jsonb	No	'{}'::jsonb	-	-	Additional structured values
created_at	timestamptz	No	now()	-	-	Event timestamp

cod_collections [MVP]

Cash-on-delivery collection and reconciliation record.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Collection ID
order_id	uuid	No	-	FK	orders.id	COD order
delivery_task_id	uuid	No	-	FK	delivery_tasks.id	Delivery task
captain_id	uuid	No	-	FK	captain_profiles.user_id	Captain
amount_paise	bigint	No	-	CHECK	-	Cash collected
status	text	No	'PENDING_COLLECTION'	CHECK	-	PENDING_COLLECTION, COLLECTED, DEPOSIT_PENDING, DEPOSITED, RECONCILED, DISPUTED
collected_at	timestamptz	Yes	-	-	-	Collection timestamp
deposited_at	timestamptz	Yes	-	-	-	Deposit timestamp
reconciled_by	uuid	Yes	-	FK	profiles.id	Finance admin
reconciled_at	timestamptz	Yes	-	-	-	Reconciliation timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (order_id)

captain_earnings [MVP]

Per-task captain earning breakdown.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Earning ID
captain_id	uuid	No	-	FK	captain_profiles.user_id	Captain
delivery_task_id	uuid	No	-	FK	delivery_tasks.id	Task
base_fare_paise	bigint	No	0	-	-	Base fare
distance_fare_paise	bigint	No	0	-	-	Distance component
waiting_fee_paise	bigint	No	0	-	-	Merchant/customer waiting
peak_incentive_paise	bigint	No	0	-	-	Peak incentive
other_incentive_paise	bigint	No	0	-	-	Other incentive
tip_paise	bigint	No	0	-	-	Customer tip
penalty_paise	bigint	No	0	-	-	Penalty deduction
total_paise	bigint	No	0	-	-	Net earning
status	text	No	'PENDING'	CHECK	-	PENDING, AVAILABLE, INCLUDED_IN_PAYOUT, PAID, REVERSED
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (delivery_task_id)

Group Style

9 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

style_groups [MVP-lite]

Occasion-planning group for wedding, party, trip, festival, get-together and custom events.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Group ID
creator_customer_id	uuid	No	-	FK	customer_profiles.user_id	Group creator/admin
name	text	No	-	-	-	Group name
occasion_type	text	No	-	CHECK	-	WEDDING, PARTY, TRIP, FESTIVAL, GET_TOGETHER, COLLEGE_EVENT, OFFICE_EVENT, PHOTOSHOOT, CUSTOM
custom_occasion_name	text	Yes	-	-	-	Custom label
event_start_date	date	Yes	-	-	-	Start date
event_end_date	date	Yes	-	-	-	End date
location_name	text	Yes	-	-	-	Event/destination name
event_location	geography(Point, 4326)	Yes	-	-	-	Optional map point
budget_min_paise	bigint	Yes	-	CHECK	-	Per-person minimum
budget_max_paise	bigint	Yes	-	CHECK	-	Per-person maximum

Attribute	Type	Null?	Default	Key	Reference	Purpose
coordination_mode	text	No	'COORDINATED_LOOKS'	CHECK	-	SAME_OUTFIT, SAME_COLOUR, SAME_COLOUR_FAMILY, SAME_THEME, MATCHING_ACCESSORIES, COORDINATED_LOOKS
include_footwear	boolean	No	true	-	-	Include footwear
include_accessories	boolean	No	true	-	-	Include accessories
status	text	No	'PLANNING'	CHECK	-	PLANNING, VOTING, FINALIZED, ORDERING, COMPLETED, ARCHIVED
invite_code	text	No	-	UNIQUE	-	Shareable join code
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

style_group_members [MVP-lite]

Invited and joined participants, including event roles such as bride, groom, friends or family.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Membership ID
group_id	uuid	No	-	FK	style_groups.id	Group
customer_id	uuid	Yes	-	FK	customer_profiles.user_id	Joined user
invited_phone	text	Yes	-	-	-	Pending invite contact
display_name	text	No	-	-	-	Member display name
event_role	text	Yes	-	-	-	Bride, groom, bridesmaid, friend, family, etc.
membership_role	text	No	'MEMBER'	CHECK	-	ADMIN or MEMBER
status	text	No	'INVITED'	CHECK	-	INVITED, JOINED, DECLINED, REMOVED, LEFT
joined_at	timestamptz	Yes	-	-	-	Join timestamp
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (group_id, customer_id)

style_group_member_preferences [Phase 2]

Private size, footwear, budget and style preferences for one group member.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Preference ID
group_member_id	uuid	No	-	FK	style_group_members.id	Member
clothing_size	text	Yes	-	-	-	Usual clothing size
footwear_size	text	Yes	-	-	-	Footwear size
preferred_colours	text[]	No	'{}'	-	-	Preferred colours
avoided_colours	text[]	No	'{}'	-	-	Avoided colours
preferred_styles	text[]	No	'{}'	-	-	Style tags
budget_min_paise	bigint	Yes	-	CHECK	-	Personal minimum
budget_max_paise	bigint	Yes	-	CHECK	-	Personal maximum
include_footwear	boolean	No	true	-	-	Footwear opt-in
include_accessories	boolean	No	true	-	-	Accessories opt-in

Attribute	Type	Null?	Default	Key	Reference	Purpose
private_body_profile_id	uuid	Yes	-	FK	customer_body_profiles.id	Private fit profile, never shown to other members
updated_at	timestamptz	No	now()	-	-	Update timestamp

Additional constraints: unique (group_member_id)

style_themes [MVP]

Colour palette, visual style, material and formality rules selected by a group.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Theme ID
group_id	uuid	No	-	FK	style_groups.id	Group
name	text	No	-	-	-	Theme name
primary_colours	text[]	No	'{}'	-	-	Main palette
secondary_colours	text[]	No	'{}'	-	-	Supporting palette
style_tags	text[]	No	'{}'	-	-	Retro, beach, traditional, minimal, etc.
material_preferences	text[]	No	'{}'	-	-	Silk, cotton, linen, etc.
formality_level	text	Yes	-	-	-	CASUAL, SMART_CASUAL, FORMAL, FESTIVE, LUXURY
weather_context	jsonb	No	'{}'::jsonb	-	-	Temperature/rain/activity context
created_by	uuid	No	-	FK	profiles.id	Creator
is_selected	boolean	No	false	-	-	Active/final theme
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

style_polls [MVP]

Group polls for colours, themes, products, merchants or budgets.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Poll ID
group_id	uuid	No	-	FK	style_groups.id	Group
question	text	No	-	-	-	Poll question
poll_type	text	No	-	CHECK	-	COLOUR, THEME, PRODUCT, MERCHANT, BUDGET, CUSTOM
allows_multiple	boolean	No	false	-	-	Multi-select flag
closes_at	timestamptz	Yes	-	-	-	Voting deadline
status	text	No	'OPEN'	CHECK	-	OPEN, CLOSED, CANCELLED
created_by	uuid	No	-	FK	profiles.id	Creator
created_at	timestamptz	No	now()	-	-	Creation timestamp

style_poll_options [MVP]

Selectable poll options that can reference a theme, product or free text.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Option ID
poll_id	uuid	No	-	FK	style_polls.id	Poll
label	text	No	-	-	-	Display label
product_id	uuid	Yes	-	FK	products.id	Product option
theme_id	uuid	Yes	-	FK	style_themes.id	Theme option
image_path	text	Yes	-	-	-	Optional preview
metadata	jsonb	No	'{}'::jsonb	-	-	Colour/budget/other data
display_order	integer	No	0	-	-	Sort order
created_at	timestamptz	No	now()	-	-	Creation timestamp

style_poll_votes [MVP]

One member vote per poll option, constrained by poll rules.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Vote ID
poll_id	uuid	No	-	FK	style_polls.id	Poll
option_id	uuid	No	-	FK	style_poll_options.id	Option
group_member_id	uuid	No	-	FK	style_group_members.id	Member
created_at	timestamptz	No	now()	-	-	Vote timestamp

Additional constraints: unique (poll_id, option_id, group_member_id)

style_moodboards [Phase 2]

Shareable group mood board containing colour palette, member looks and accessories.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Moodboard ID
group_id	uuid	No	-	FK	style_groups.id	Group
theme_id	uuid	Yes	-	FK	style_themes.id	Theme
name	text	No	-	-	-	Moodboard name
cover_image_path	text	Yes	-	-	-	Generated/exported board
status	text	No	'DRAFT'	CHECK	-	DRAFT, SHARED, FINAL
created_by	uuid	No	-	FK	profiles.id	Creator
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

style_moodboard_items [Phase 2]

Products, palettes and member recommendation cards positioned on a mood board.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Moodboard item ID
moodboard_id	uuid	No	-	FK	style_moodboards.id	Moodboard
group_member_id	uuid	Yes	-	FK	style_group_members.id	Assigned member
product_id	uuid	Yes	-	FK	products.id	Product
variant_id	uuid	Yes	-	FK	product_variants.id	Variant
item_type	text	No	-	CHECK	-	PRODUCT, COLOUR_SWATCH, TEXT, IMAGE
item_role	text	Yes	-	-	-	MAIN_CLOTHING, FOOTWEAR, BELT, WATCH, BAG, ACCESSORY
position_data	jsonb	No	'{}'::jsonb	-	-	UI placement
created_at	timestampz	No	now()	-	-	Creation timestamp

Engagement

3 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

shop_favourites [MVP]

Customer-to-shop follow/favourite relationship created voluntarily after search, visit or merchant request.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Favourite ID
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
shop_id	uuid	No	-	FK	shops.id	Followed shop
source	text	No	'SHOP_PAGE'	CHECK	-	ORGANIC_SEARCH, SHOP_PAGE, MERCHANT_REQUEST, ORDER_HISTORY, PRODUCT_PAGE, CAMPAIGN
new_arrival_notifications	boolean	No	true	-	-	New-arrival updates
offer_notifications	boolean	No	true	-	-	Shop offers
restock_notifications	boolean	No	true	-	-	Restock alerts
favourited_at	timestampz	No	now()	-	-	Follow timestamp
unfavourited_at	timestampz	Yes	-	-	-	Soft unfollow timestamp

Additional constraints: unique (customer_id, shop_id)

wishlist_items [MVP]

Customer product/variant wishlist.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Wishlist row
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
product_id	uuid	No	-	FK	products.id	Product

Attribute	Type	Null?	Default	Key	Reference	Purpose
variant_id	uuid	Yes	-	FK	product_variants.id	Optional selected variant
created_at	timestamptz	No	now()	-	-	Save timestamp

Additional constraints: unique (customer_id, product_id, variant_id)

reviews [MVP]

Verified-order reviews for product, shop and captain.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Review ID
order_id	uuid	No	-	FK	orders.id	Verified order
customer_id	uuid	No	-	FK	customer_profiles.user_id	Reviewer
review_type	text	No	-	CHECK	-	PRODUCT, SHOP, CAPTAIN
shop_id	uuid	Yes	-	FK	shops.id	Shop target
product_id	uuid	Yes	-	FK	products.id	Product target
captain_id	uuid	Yes	-	FK	captain_profiles.user_id	Captain target
rating	smallint	No	-	CHECK	-	1-5
review_text	text	Yes	-	-	-	Review
moderation_status	text	No	'PUBLISHED'	CHECK	-	PUBLISHED, PENDING, HIDDEN, REJECTED
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

Notifications

3 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

notifications [MVP]

In-app notification record for customer, merchant, captain and admin users.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Notification ID
user_id	uuid	No	-	FK	profiles.id	Recipient
notification_type	text	No	-	-	-	ORDER, DELIVERY, INVENTORY, OFFER, GROUP, SUPPORT, PAYOUT, SYSTEM
title	text	No	-	-	-	Title
body	text	No	-	-	-	Message
entity_type	text	Yes	-	-	-	Related table/entity type
entity_id	uuid	Yes	-	-	-	Related ID
priority	text	No	'NORMAL'	CHECK	-	LOW, NORMAL, HIGH, URGENT
data	jsonb	No	'{}'::jsonb	-	-	Deep-link/action data
created_at	timestamptz	No	now()	-	-	Creation timestamp
read_at	timestamptz	Yes	-	-	-	Read timestamp

notification_deliveries [MVP]

Channel-specific send, retry and delivery receipt history.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Delivery attempt sequence
notification_id	uuid	No	-	FK	notifications.id	Notification
device_id	uuid	Yes	-	FK	user_devices.id	Device for push
channel	text	No	-	CHECK	-	PUSH, SMS, WHATSAPP, EMAIL, IN_APP
provider	text	Yes	-	-	-	FCM, APNS, MSG91, etc.
provider_message_id	text	Yes	-	-	-	Provider identifier
status	text	No	'PENDING'	CHECK	-	PENDING, SENT, DELIVERED, READ, FAILED
attempt_count	smallint	No	0	CHECK	-	Attempts
sent_at	timestamptz	Yes	-	-	-	Sent
received_at	timestamptz	Yes	-	-	-	Delivered/receipt
failed_reason	text	Yes	-	-	-	Failure detail
created_at	timestamptz	No	now()	-	-	Creation timestamp

notification_preferences [MVP]

Global non-transactional notification preferences. Critical order and security alerts are controlled separately.

Attribute	Type	Null?	Default	Key	Reference	Purpose
user_id	uuid	No	-	PK/FK	profiles.id	User
order_updates	boolean	No	true	-	-	Transactional order updates
delivery_updates	boolean	No	true	-	-	Delivery updates
shop_offers	boolean	No	true	-	-	Favourite-shop offers
new_arrivals	boolean	No	true	-	-	Favourite-shop arrivals
recommendations	boolean	No	true	-	-	Personal recommendations
group_updates	boolean	No	true	-	-	Group Style activity
support_updates	boolean	No	true	-	-	Ticket status
quiet_hours_start	time	Yes	-	-	-	Marketing quiet hours
quiet_hours_end	time	Yes	-	-	-	Marketing quiet hours
updated_at	timestamptz	No	now()	-	-	Update timestamp

Promotions

3 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

offers [MVP]

Merchant-created discounts and bundle/free-delivery offers.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Offer ID
shop_id	uuid	No	-	FK	shops.id	Shop
name	text	No	-	-	-	Offer name
description	text	Yes	-	-	-	Customer-facing details
offer_type	text	No	-	CHECK	-	PERCENT, FIXED, FREE_DELIVERY, BUY_X_GET_Y, BUNDLE
discount_value	numeric(12,2)	Yes	-	CHECK	-	Percent or rupee value depending on type
minimum_order_paise	bigint	No	0	CHECK	-	Minimum eligible order
maximum_discount_paise	bigint	Yes	-	CHECK	-	Discount cap
usage_limit_total	integer	Yes	-	CHECK	-	Total redemptions
usage_limit_per_customer	integer	No	1	CHECK	-	Per-customer cap
starts_at	timestamptz	No	-	-	-	Start
ends_at	timestamptz	No	-	-	-	End
status	text	No	'DRAFT'	CHECK	-	DRAFT, ACTIVE, PAUSED, EXPIRED
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

offer_products [MVP]

Products/variants explicitly included in an offer.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Mapping ID
offer_id	uuid	No	-	FK	offers.id	Offer
product_id	uuid	No	-	FK	products.id	Product
variant_id	uuid	Yes	-	FK	product_variants.id	Optional variant
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (offer_id, product_id, variant_id)

offer_redemptions [MVP]

Immutable application of an offer to a completed/placed order.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Redemption ID
offer_id	uuid	No	-	FK	offers.id	Offer
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
order_id	uuid	No	-	FK	orders.id	Order
discount_paise	bigint	No	-	CHECK	-	Discount applied

Attribute	Type	Null?	Default	Key	Reference	Purpose
created_at	timestamptz	No	now()	-	-	Redemption timestamp

Additional constraints: unique (offer_id, customer_id, order_id)

Campaigns

2 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

campaigns [MVP]

Admin-created banners, coupons, city promotions and Group Style campaigns.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Campaign ID
name	text	No	-	-	-	Campaign name
campaign_type	text	No	-	CHECK	-	BANNER, COUPON, FREE_DELIVERY, FESTIVAL, GROUP_STYLE, MERCHANT_SPONSORED
objective	text	Yes	-	-	-	Acquisition, conversion, retention, etc.
audience_rules	jsonb	No	'{}':jsonb	-	-	City, segment, favourite-shop, category targeting
budget_paise	bigint	Yes	-	CHECK	-	Optional budget
status	text	No	'DRAFT'	CHECK	-	DRAFT, APPROVAL_REQUIRED, SCHEDULED, ACTIVE, PAUSED, COMPLETED, CANCELLED
starts_at	timestamptz	No	-	-	-	Start
ends_at	timestamptz	No	-	-	-	End
created_by	uuid	No	-	FK	profiles.id	Marketing admin
approved_by	uuid	Yes	-	FK	profiles.id	Approver
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

banners [MVP]

Responsive banner creative and destination shown on customer home/discovery surfaces.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Banner ID
campaign_id	uuid	No	-	FK	campaigns.id	Campaign
mobile_image_path	text	No	-	-	-	Mobile creative
desktop_image_path	text	Yes	-	-	-	Web creative
title	text	No	-	-	-	Banner title
subtitle	text	Yes	-	-	-	Supporting text
cta_text	text	Yes	-	-	-	Call to action
destination_type	text	No	-	CHECK	-	CATEGORY, PRODUCT, SHOP, COLLECTION, GROUP_STYLE, OFFER, URL
destination_id	uuid	Yes	-	-	-	Internal entity ID

Attribute	Type	Null?	Default	Key	Reference	Purpose
external_url	text	Yes	-	-	-	Approved external destination
priority	integer	No	0	-	-	Sort priority
placement	text	No	'HOME_TOP'	-	-	HOME_TOP, HOME_MID, CATEGORY, SEARCH
impression_count	bigint	No	0	-	-	Cached analytics
click_count	bigint	No	0	-	-	Cached analytics
created_at	timestamptz	No	now()	-	-	Creation timestamp

Intelligence

16 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

customer_events [MVP-data]

High-volume behavioural event stream for recommendation training and analytics.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Event sequence
customer_id	uuid	Yes	-	FK	customer_profiles.user_id	Signed-in customer
anonymous_id	uuid	Yes	-	-	-	Pre-login device/session identity
session_id	uuid	No	-	-	-	Session
app_name	text	No	'CUSTOMER'	-	-	Source app
event_type	text	No	-	CHECK	-	PRODUCT_VIEW, PRODUCT_CLICK, SEARCH, SHOP_VIEW, FAVOURITE_SHOP, WISHLIST_ADD, CART_ADD, PURCHASE, RETURN, GROUP_SELECTION, FIT_FEEDBACK
entity_type	text	Yes	-	-	-	PRODUCT, SHOP, CATEGORY, OFFER, GROUP, etc.
entity_id	uuid	Yes	-	-	-	Related entity
shop_id	uuid	Yes	-	FK	shops.id	Shop context
query_text	text	Yes	-	-	-	Search query when applicable
context	jsonb	No	'{}'::jsonb	-	-	Filters, rank, source and device context
occurred_at	timestamptz	No	now()	-	-	Client event time
received_at	timestamptz	No	now()	-	-	Server time

recommendation_sets [MVP-rules]

One generated ranking for a customer and context.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Recommendation set ID
customer_id	uuid	Yes	-	FK	customer_profiles.user_id	Customer
session_id	uuid	Yes	-	-	-	Session-based ranking

Attribute	Type	Null?	Default	Key	Reference	Purpose
recommendation_type	text	No	-	CHECK	-	HOME_PERSONALIZED, SIMILAR_PRODUCTS, COMPLETE_THE_LOOK, TRENDING_NEARBY, BASED_ON_OCCASION, BASED_ON_FAVOURITE_SHOP, GROUP_STYLE
context	jsonb	No	'{}'::jsonb	-	-	Location, source product, occasion, group, etc.
algorithm_version	text	No	'rules-v1'	-	-	Rules/model version
generated_at	timestamptz	No	now()	-	-	Generation timestamp
expires_at	timestamptz	Yes	-	-	-	Cache expiry

recommendation_items [MVP-rules]

Ranked recommendation output with explanation and engagement timestamps.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Recommendation item ID
recommendation_set_id	uuid	No	-	FK	recommendation_sets.id	Set
product_id	uuid	No	-	FK	products.id	Product
variant_id	uuid	Yes	-	FK	product_variants.id	Optional exact variant
rank_position	integer	No	-	CHECK	-	Rank
score	numeric(10,8)	No	-	-	-	Ranking score
reason_codes	text[]	No	'{}'	-	-	Explainable reasons
score_components	jsonb	No	'{}'::jsonb	-	-	Category/style/colour/budget/distance scores
shown_at	timestamptz	Yes	-	-	-	Impression
clicked_at	timestamptz	Yes	-	-	-	Click
purchased_at	timestamptz	Yes	-	-	-	Conversion
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (recommendation_set_id, rank_position)

product_embeddings [Phase 3]

Vector representation for image/text similarity and outfit compatibility.

Attribute	Type	Null?	Default	Key	Reference	Purpose
product_id	uuid	No	-	PK/FK	products.id	Product
embedding_type	text	No	'MULTIMODAL'	-	-	TEXT, IMAGE or MULTIMODAL
embedding	vector(768)	No	-	-	-	Vector, dimension may change by model
model_version	text	No	-	-	-	Embedding model version
source_hash	text	No	-	-	-	Hash of content used to detect staleness
updated_at	timestamptz	No	now()	-	-	Update timestamp

customer_embeddings [Phase 3]

Learned customer preference vector for ML ranking.

Attribute	Type	Null?	Default	Key	Reference	Purpose
customer_id	uuid	No	-	PK/FK	customer_profiles.user_id	Customer
embedding	vector(768)	No	-	-	-	Preference vector
model_version	text	No	-	-	-	Model version
source_event_count	integer	No	0	CHECK	-	Events used
updated_at	timestamptz	No	now()	-	-	Update timestamp

size_charts [MVP-size]

Brand/shop/category size-chart version.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Size chart ID
shop_id	uuid	Yes	-	FK	shops.id	Shop-specific chart
brand	text	Yes	-	-	-	Brand
category_id	uuid	No	-	FK	categories.id	Category
gender_category	text	No	'UNISEX'	-	-	Target category
country_standard	text	No	'IN'	-	-	Sizing standard
name	text	No	-	-	-	Chart name
version	integer	No	1	CHECK	-	Version
is_active	boolean	No	true	-	-	Active chart
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

size_chart_measurements [MVP-size]

Measurement ranges and garment dimensions for each label in a size chart.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Measurement row
size_chart_id	uuid	No	-	FK	size_charts.id	Chart
size_label	text	No	-	-	-	S, M, 38, footwear 9, etc.
measurement_name	text	No	-	CHECK	-	HEIGHT, CHEST, WAIST, HIP, SHOULDER, SLEEVE, TORSO, INSEAM, FOOT_LENGTH, GARMENT_LENGTH
min_cm	numeric(8,2)	Yes	-	CHECK	-	Body minimum
max_cm	numeric(8,2)	Yes	-	CHECK	-	Body maximum
garment_measurement_cm	numeric(8,2)	Yes	-	CHECK	-	Actual garment measurement
tolerance_cm	numeric(8,2)	Yes	-	CHECK	-	Ease/tolerance
created_at	timestamptz	No	now()	-	-	Creation timestamp

Additional constraints: unique (size_chart_id, size_label, measurement_name)

customer_body_profiles [Phase 2]

Consent-controlled body profile for manual measurements, camera scan or learned fit history.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Body profile ID
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
profile_name	text	No	'My Fit Profile'	-	-	Customer label
source	text	No	'MANUAL_ENTRY'	CHECK	-	MANUAL_ENTRY, CAMERA_SCAN, PURCHASE_HISTORY
fit_preference	text	No	'REGULAR'	CHECK	-	SLIM, REGULAR, RELAXED, OVERSIZED
height_cm	numeric(8,2)	Yes	-	CHECK	-	Height input/reference
weight_kg	numeric(8,2)	Yes	-	CHECK	-	Weight input/reference
consent_version	text	No	-	-	-	Privacy consent version
consent_given_at	timestamptz	No	-	-	-	Consent timestamp
is_active	boolean	No	true	-	-	Active profile
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp
deleted_at	timestamptz	Yes	-	-	-	Customer deletion request

body_measurements [Phase 2]

Individual measurements with confidence and source.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Measurement ID
body_profile_id	uuid	No	-	FK	customer_body_profiles.id	Body profile
measurement_name	text	No	-	CHECK	-	CHEST, WAIST, HIP, SHOULDER, ARM_LENGTH, TORSO_LENGTH, LEG_LENGTH, INSEAM, FOOT_LENGTH
value_cm	numeric(8,2)	No	-	CHECK	-	Measurement
confidence_score	numeric(6,5)	Yes	-	CHECK	-	Confidence for estimated values
source	text	No	-	CHECK	-	USER_ENTERED, CAMERA_ESTIMATED, PURCHASE_LEARNED
captured_at	timestamptz	No	now()	-	-	Capture timestamp

Additional constraints: unique (body_profile_id, measurement_name)

body_scan_sessions [Phase 3]

Privacy-sensitive guided body scan processing session. Raw images should be deleted by default after extraction.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Scan session ID
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
body_profile_id	uuid	Yes	-	FK	customer_body_profiles.id	Result profile
status	text	No	'CREATED'	CHECK	-	CREATED, UPLOADING, QUALITY_CHECK, PROCESSING, COMPLETED, FAILED, DELETED
front_image_path	text	Yes	-	-	-	Private temporary image

Attribute	Type	Null?	Default	Key	Reference	Purpose
side_image_path	text	Yes	-	-	-	Private temporary image
back_image_path	text	Yes	-	-	-	Optional private image
processing_provider	text	Yes	-	-	-	Internal/provider
model_version	text	Yes	-	-	-	Model
quality_score	numeric(6,5)	Yes	-	CHECK	-	Scan quality
quality_issues	text[]	No	'{}'	-	-	Full body missing, lighting, tilt, etc.
error_message	text	Yes	-	-	-	Failure detail
deletion_scheduled_at	timestamptz	Yes	-	-	-	Raw-image deletion time
created_at	timestamptz	No	now()	-	-	Creation timestamp
completed_at	timestamptz	Yes	-	-	-	Completion timestamp

fit_recommendations [MVP-size]

Per-product size recommendation, confidence, explanation and model version.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Fit recommendation ID
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
body_profile_id	uuid	Yes	-	FK	customer_body_profiles.id	Body profile
product_id	uuid	No	-	FK	products.id	Product
variant_id	uuid	Yes	-	FK	product_variants.id	Recommended variant
recommended_size	text	No	-	-	-	Suggested size
confidence_score	numeric(6,5)	No	-	CHECK	-	Confidence
fit_prediction	text	No	'EXPECTED_FIT'	CHECK	-	TOO_SMALL, SLIGHTLY_SMALL, EXPECTED_FIT, SLIGHTLY_LARGE, TOO_LARGE
reason_codes	text[]	No	'{}'	-	-	Explainable reasons
inputs_snapshot	jsonb	No	'{}':jsonb	-	-	Measurements and chart version used
model_version	text	No	'rules-v1'	-	-	Model/rules version
created_at	timestamptz	No	now()	-	-	Creation timestamp

fit_feedback [Phase 2]

Post-delivery fit outcome used to improve brand/product sizing.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Feedback ID
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
order_item_id	uuid	No	-	FK	order_items.id	Purchased item
fit_recommendation_id	uuid	Yes	-	FK	fit_recommendations.id	Prior prediction
predicted_size	text	Yes	-	-	-	Predicted
purchased_size	text	No	-	-	-	Purchased
feedback	text	No	-	CHECK	-	TOO_SMALL, SLIGHTLY_SMALL, PERFECT, SLIGHTLY_LARGE, TOO_LARGE

Attribute	Type	Null?	Default	Key	Reference	Purpose
returned_due_to_fit	boolean	No	false	-	-	Fit return marker
created_at	timestamptz	No	now()	-	-	Feedback timestamp

Additional constraints: unique (order_item_id)

virtual_tryon_sessions [Phase 3]

Photo/avatar virtual try-on job with provider/model, consent and retention state.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Try-on session ID
customer_id	uuid	No	-	FK	customer_profiles.user_id	Customer
product_id	uuid	No	-	FK	products.id	Garment
variant_id	uuid	Yes	-	FK	product_variants.id	Colour/variant
input_type	text	No	-	CHECK	-	UPLOADED_PHOTO, SAVED_BODY_PROFILE, GENERIC_MODEL
customer_image_path	text	Yes	-	-	-	Private input photo
body_profile_id	uuid	Yes	-	FK	customer_body_profiles.id	Avatar/body profile
status	text	No	'CREATED'	CHECK	-	CREATED, PROCESSING, COMPLETED, FAILED, DELETED
provider	text	Yes	-	-	-	Processing provider
model_version	text	Yes	-	-	-	Model version
consent_version	text	No	-	-	-	Consent version
deletion_scheduled_at	timestamptz	Yes	-	-	-	Input/output deletion policy
error_message	text	Yes	-	-	-	Failure detail
created_at	timestamptz	No	now()	-	-	Creation timestamp
completed_at	timestamptz	Yes	-	-	-	Completion timestamp

virtual_tryon_outputs [Phase 3]

Generated front/side/back try-on output references.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Output ID
tryon_session_id	uuid	No	-	FK	virtual_tryon_sessions.id	Session
view_type	text	No	'FRONT'	CHECK	-	FRONT, SIDE, BACK, GROUP_MOODBOARD
output_image_path	text	No	-	-	-	Private generated image
quality_score	numeric(6,5)	Yes	-	CHECK	-	Output quality
is_saved_by_customer	boolean	No	false	-	-	Customer save state
created_at	timestamptz	No	now()	-	-	Creation timestamp

style_recommendation_sets [Phase 2]

Scored coordinated set for a group theme and members.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Set ID

Attribute	Type	Null?	Default	Key	Reference	Purpose
group_id	uuid	No	-	FK	style_groups.id	Group
theme_id	uuid	Yes	-	FK	style_themes.id	Theme
algorithm_version	text	No	'rules-v1'	-	-	Ranking version
total_score	numeric(10,8)	No	0	-	-	Combined score
score_components	jsonb	No	'{}':jsonb	-	-	Theme, stock, budget, delivery scores
status	text	No	'GENERATED'	CHECK	-	GENERATED, SHORTLISTED, SELECTED, EXPIRED
created_at	timestampz	No	now()	-	-	Generation timestamp

style_recommendation_items [Phase 2]

Member-level clothing, footwear and accessory products in a coordinated group set.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Item ID
recommendation_set_id	uuid	No	-	FK	style_recommendation_sets.id	Set
group_member_id	uuid	No	-	FK	style_group_members.id	Member
product_id	uuid	No	-	FK	products.id	Product
variant_id	uuid	Yes	-	FK	product_variants.id	Suggested variant
item_role	text	No	-	CHECK	-	MAIN_CLOTHING, BOTTOM_WEAR, OUTERWEAR, FOOTWEAR, BELT, WATCH, BAG, JEWELLERY, SCARF, POCKET_SQUARE, OTHER_ACCESSORY
compatibility_score	numeric(10,8)	No	0	-	-	Style compatibility
availability_score	numeric(10,8)	No	0	-	-	Stock/delivery score
reason_codes	text[]	No	'{}'	-	-	Explanation
created_at	timestampz	No	now()	-	-	Creation timestamp

Operations

2 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

support_tickets [MVP]

Unified support ticket for customer, merchant, captain and internal issues.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Ticket ID
ticket_number	text	No	-	UNIQUE	-	Human-friendly ticket number
raised_by_user_id	uuid	No	-	FK	profiles.id	Requester
raised_by_type	text	No	-	CHECK	-	CUSTOMER, MERCHANT, CAPTAIN, ADMIN
order_id	uuid	Yes	-	FK	orders.id	Related order
shop_id	uuid	Yes	-	FK	shops.id	Related shop
delivery_task_id	uuid	Yes	-	FK	delivery_tasks.id	Related delivery

Attribute	Type	Null?	Default	Key	Reference	Purpose
return_request_id	uuid	Yes	-	FK	return_requests.id	Related return
category	text	No	-	-	-	ORDER, PAYMENT, INVENTORY, PRODUCT, DELIVERY, RETURN, ACCOUNT, TECHNICAL, SAFETY, OTHER
priority	text	No	'MEDIUM'	CHECK	-	LOW, MEDIUM, HIGH, URGENT
status	text	No	'OPEN'	CHECK	-	OPEN, ASSIGNED, IN_PROGRESS, WAITING_FOR_USER, ESCALATED, RESOLVED, CLOSED
subject	text	No	-	-	-	Short description
description	text	No	-	-	-	Full description
assigned_team	text	Yes	-	-	-	Support queue
assigned_to	uuid	Yes	-	FK	profiles.id	Agent
resolution_code	text	Yes	-	-	-	Resolution category
resolution_note	text	Yes	-	-	-	Resolution
created_at	timestamptz	No	now()	-	-	Creation timestamp
first_response_at	timestamptz	Yes	-	-	-	SLA response
resolved_at	timestamptz	Yes	-	-	-	Resolution
closed_at	timestamptz	Yes	-	-	-	Closure
updated_at	timestamptz	No	now()	-	-	Update timestamp

support_messages [MVP]

Ticket conversation, attachments and internal notes.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Message sequence
ticket_id	uuid	No	-	FK	support_tickets.id	Ticket
sender_id	uuid	No	-	FK	profiles.id	Sender
message_type	text	No	'TEXT'	CHECK	-	TEXT, IMAGE, FILE, SYSTEM
message	text	Yes	-	-	-	Message content
attachment_path	text	Yes	-	-	-	Private Storage attachment
is_internal_note	boolean	No	false	-	-	Invisible to requester
created_at	timestamptz	No	now()	-	-	Creation timestamp

Field Operations

3 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

merchant_leads [MVP]

Prospective shop pipeline for field acquisition and onboarding.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Lead ID

Attribute	Type	Null?	Default	Key	Reference	Purpose
shop_name	text	No	-	-	-	Shop name
owner_name	text	Yes	-	-	-	Owner
phone_number	text	No	-	-	-	Contact
category_summary	text	Yes	-	-	-	Main inventory categories
address_text	text	Yes	-	-	-	Address
location	geography(Point, 4326)	Yes	-	-	-	Shop point
lead_source	text	No	'FIELD'	-	-	FIELD, CUSTOMER_REQUEST, REFERRAL, INBOUND
lead_status	text	No	'NEW'	CHECK	-	NEW, CONTACTED, INTERESTED, VISIT_SCHEDULED, ONBOARDING_STARTED, ACTIVE, NOT_INTERESTED, FOLLOW_UP
assigned_executive_id	uuid	Yes	-	FK	profiles.id	Field executive
next_followup_at	timestamptz	Yes	-	-	-	Follow-up
notes	text	Yes	-	-	-	Notes
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

field_visits [MVP]

GPS and merchant-confirmed merchant onboarding/support visit.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Visit ID
merchant_lead_id	uuid	Yes	-	FK	merchant_leads.id	Lead
shop_id	uuid	Yes	-	FK	shops.id	Existing shop
executive_id	uuid	No	-	FK	profiles.id	Assigned executive
visit_type	text	No	-	CHECK	-	SALES, ONBOARDING, TRAINING, SUPPORT, INVENTORY_AUDIT, RETENTION
scheduled_at	timestamptz	No	-	-	-	Schedule
started_at	timestamptz	Yes	-	-	-	Start
completed_at	timestamptz	Yes	-	-	-	Completion
start_location	geography(Point, 4326)	Yes	-	-	-	GPS verification
completion_location	geography(Point, 4326)	Yes	-	-	-	GPS verification
merchant_confirmation_method	text	Yes	-	CHECK	-	OTP, SIGNATURE, APP_CONFIRMATION
merchant_confirmation_at	timestamptz	Yes	-	-	-	Confirmation
status	text	No	'SCHEDULED'	CHECK	-	SCHEDULED, IN_PROGRESS, COMPLETED, CANCELLED, RESCHEDULED
notes	text	Yes	-	-	-	Visit notes
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

field_visit_checklist_items [MVP]

Structured onboarding/training/support checklist evidence.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Checklist item ID
field_visit_id	uuid	No	-	FK	field_visits.id	Visit
item_code	text	No	-	-	-	SHOP_PHOTO, KYC, FIRST_PRODUCT, INVENTORY, TEST_ORDER, RINGTONE_TEST, TRAINING, etc.
label	text	No	-	-	-	Display label
is_required	boolean	No	true	-	-	Required before completion
is_completed	boolean	No	false	-	-	Completion
evidence_path	text	Yes	-	-	-	Optional evidence
completed_by	uuid	Yes	-	FK	profiles.id	Executive
completed_at	timestamptz	Yes	-	-	-	Completion timestamp
notes	text	Yes	-	-	-	Notes

Additional constraints: unique (field_visit_id, item_code)

Moderation

1 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

product_moderation_cases [MVP]

Product quality, duplicate, category, pricing and prohibited-content moderation case.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Case ID
product_id	uuid	No	-	FK	products.id	Product
case_type	text	No	-	CHECK	-	NEW_LISTING, IMAGE_QUALITY, WRONG_CATEGORY, DUPLICATE, MISLEADING_PRICE, PROHIBITED, CUSTOMER_REPORT
status	text	No	'OPEN'	CHECK	-	OPEN, ASSIGNED, MERCHANT_ACTION_REQUIRED, APPROVED, REJECTED, CLOSED
moderator_id	uuid	Yes	-	FK	profiles.id	Moderator
merchant_response	text	Yes	-	-	-	Merchant response
resolution_note	text	Yes	-	-	-	Resolution
created_at	timestamptz	No	now()	-	-	Creation timestamp
resolved_at	timestamptz	Yes	-	-	-	Resolution timestamp

Risk

1 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

fraud_cases [Phase 2]

Investigation record for suspicious orders, referrals, refunds, COD, merchants or captains.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Fraud case ID
case_number	text	No	-	UNIQUE	-	Human-friendly case number
case_type	text	No	-	CHECK	-	FAKE_ORDER, DUPLICATE_ACCOUNT, REFUND_ABUSE, COD_ABUSE, MERCHANT_MANIPULATION, CAPTAIN_FRAUD, ACCOUNT_TAKEOVER, OTHER
subject_user_id	uuid	Yes	-	FK	profiles.id	Subject user
shop_id	uuid	Yes	-	FK	shops.id	Subject shop
order_id	uuid	Yes	-	FK	orders.id	Subject order
risk_score	numeric(6,3)	Yes	-	CHECK	-	Risk score
signals	jsonb	No	'{}'::jsonb	-	-	Detection signals
status	text	No	'OPEN'	CHECK	-	OPEN, INVESTIGATING, ACTION_REQUIRED, CLEARED, CONFIRMED, CLOSED
assigned_to	uuid	Yes	-	FK	profiles.id	Risk analyst
action_taken	text	Yes	-	-	-	Hold, warning, block, etc.
notes	text	Yes	-	-	-	Investigation notes
created_at	timestamptz	No	now()	-	-	Creation timestamp
resolved_at	timestamptz	Yes	-	-	-	Resolution

Governance

2 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

approval_requests [MVP]

Maker-checker workflow for large refunds, bank changes, suspensions, settlement adjustments and campaigns.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Approval ID
action_type	text	No	-	CHECK	-	LARGE_REFUND, BANK_CHANGE, MERCHANT_SUSPENSION, CAPTAIN_SUSPENSION, SETTLEMENT_ADJUSTMENT, CAMPAIGN_BUDGET, OTHER
entity_type	text	No	-	-	-	Target entity
entity_id	uuid	No	-	-	-	Target ID
requested_by	uuid	No	-	FK	profiles.id	Maker
requested_changes	jsonb	No	-	-	-	Proposed mutation
reason	text	No	-	-	-	Business reason
status	text	No	'PENDING'	CHECK	-	PENDING, APPROVED, REJECTED, CANCELLED, EXECUTED
approved_by	uuid	Yes	-	FK	profiles.id	Checker
decision_note	text	Yes	-	-	-	Decision
created_at	timestamptz	No	now()	-	-	Creation timestamp

Attribute	Type	Null?	Default	Key	Reference	Purpose
decided_at	timestamptz	Yes	-	-	-	Decision timestamp
executed_at	timestamptz	Yes	-	-	-	Execution timestamp

audit_logs [MVP]

Append-only record of sensitive mutations and admin/backend actions. Partition monthly and never expose to clients.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	bigint	No	generated always as identity	PK	-	Audit sequence
actor_user_id	uuid	Yes	-	FK	profiles.id	User or null for system
actor_role	text	No	'SYSTEM'	-	-	Role/context
action	text	No	-	-	-	Stable action code
entity_type	text	No	-	-	-	Target type
entity_id	uuid	Yes	-	-	-	Target ID
old_values	jsonb	Yes	-	-	-	Sanitized before state
new_values	jsonb	Yes	-	-	-	Sanitized after state
reason	text	Yes	-	-	-	Business reason
ip_address	inet	Yes	-	-	-	Request IP
device_id	uuid	Yes	-	FK	user_devices.id	Device
request_id	uuid	Yes	-	-	-	Trace/correlation ID
created_at	timestamptz	No	now()	-	-	Immutable timestamp

Configuration

2 tables. Fields marked Phase 2/3 are reserved for later rollout but use the same Supabase project and security model.

system_settings [MVP]

Versioned platform configuration such as timeouts, limits and feature flags.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Setting ID
setting_key	text	No	-	UNIQUE	-	Stable key
setting_value	jsonb	No	-	-	-	Value
value_type	text	No	-	CHECK	-	BOOLEAN, NUMBER, STRING, JSON
scope_type	text	No	'GLOBAL'	CHECK	-	GLOBAL, CITY, SHOP
scope_id	text	Yes	-	-	-	City/shop identifier
is_secret	boolean	No	false	-	-	Secret values should normally use Vault instead
version	integer	No	1	CHECK	-	Version
updated_by	uuid	No	-	FK	profiles.id	Admin
updated_at	timestamptz	No	now()	-	-	Update timestamp

Additional constraints: unique (setting_key, scope_type, scope_id)

service_zones [MVP]

Polygonal city/service areas used for customer serviceability and operational routing.

Attribute	Type	Null?	Default	Key	Reference	Purpose
id	uuid	No	gen_random_uuid()	PK	-	Zone ID
name	text	No	-	-	-	Zone name
city	text	No	-	-	-	City
zone_type	text	No	'DELIVERY'	CHECK	-	DELIVERY, MERCHANT_ONBOARDING, CAPTAIN_OPERATIONS
boundary	geography(Polygon, 4326)	No	-	-	-	Service polygon
is_active	boolean	No	true	-	-	Active flag
base_delivery_fee_paise	bigint	No	0	CHECK	-	Base fee
per_km_fee_paise	bigint	No	0	CHECK	-	Distance fee
minimum_order_paise	bigint	No	0	CHECK	-	Zone minimum
created_at	timestamptz	No	now()	-	-	Creation timestamp
updated_at	timestamptz	No	now()	-	-	Update timestamp

Part IV - Security, Realtime, AI and operations

7. Row Level Security model

- Enable and FORCE RLS on every public table. A table with no policy is intentionally inaccessible through public/publishable clients.
- Use `auth.uid()` to bind a row to the signed-in profile, and index every ownership column used in policies.
- Use private SECURITY DEFINER helper functions for shop ownership, group membership and admin permissions; keep them outside exposed schemas.
- Use a publishable key in all apps. Secret keys bypass RLS and belong only in trusted servers or Edge Functions.
- Do not rely on user-editable `raw_user_meta_data` for authorization. Self-registration defaults to CUSTOMER; trusted workflows elevate roles.
- Test RLS with separate customer, merchant, captain, support, finance and unauthenticated test identities.

8. Storage design

Item	Details
product-images	Public read; merchant/backend uploads; optimized thumbnails.
shop-images	Public read; shop logo/cover/gallery.
merchant-documents	Private; KYC, licences, bank proof.
captain-documents	Private; identity, licence, RC, insurance.
return-evidence	Private; customer/merchant/captain evidence.
body-scans	Highly private and temporary; scheduled deletion.
virtual-tryon	Private generated inputs/outputs with retention controls.
support-attachments	Private ticket attachments.

9. Realtime and merchant ringing

Realtime is excellent for foreground screens, but it cannot guarantee a ringtone when the Merchant App is backgrounded or terminated. Use both layers:

5. Order transaction commits and creates `merchant_order_alerts`.
6. Realtime subscription shows a full in-app incoming-order modal and loops the custom sound while the app is open.
7. Backend/Edge Function sends high-priority FCM or APNs notification through a dedicated New Orders channel.
8. Retry attempts and acknowledgement are written to `merchant_order_alerts`; timeout triggers auto-rejection/escalation policy.
9. Customer and Admin see status changes through `orders/order_status_history` subscriptions.

10. Recommendation, fit and AI space

Item	Details
Rules first	customer_preferences + stock + distance + rating + budget populate recommendation_sets/items.
Behaviour later	customer_events produces co-view, co-cart, purchase and return signals.
Embeddings	product_embeddings/customer_embeddings support visual similarity and personalized ranking when data volume is sufficient.
Size fitting	size_charts + measurements + body profile + fit feedback. Always return confidence and explanation.
Camera scan	body_scan_sessions stores temporary private photos, quality checks and extracted measurements; delete raw images by default.
Virtual try-on	Asynchronous session/output tables. Treat results as visual approximation, not guaranteed fit.
Group intelligence	style themes, member preferences and availability feed coordinated recommendation sets and moodboards.

11. Performance and scaling

- GIST indexes for shops, captain locations, delivery points and service-zone polygons.
- B-tree indexes on foreign keys, status columns and recent-time queries; GIN/trigram indexes for catalogue tags and names.
- Partition customer_events, captain_location_history, notification_deliveries, payment_events, order/return history and audit_logs when volume grows.
- Keep high-frequency captain coordinates in Realtime Broadcast/Redis and persist sampled points or major events to PostgreSQL.
- Use cursor pagination for orders/products/events; avoid unbounded Realtime subscriptions.
- Generate database TypeScript types after every migration and monitor slow queries/RLS query plans.

12. Rollout sequence

Item	Details
MVP foundation	Identity/RBAC, shops, catalogue, variants, inventory, carts/orders, payments, delivery, notifications, support, returns and admin audit.
MVP differentiators	Favourite shops, Group Style basic groups/themes, rules-based recommendations and manual size charts.
Phase 2	Photo matching, full group polls/moodboards, fit feedback, advanced returns, field operations and risk.
Phase 3	Embeddings/ML ranking, camera body measurement, virtual try-on and generated group previews.

13. Official Supabase references

- API keys: <https://supabase.com/docs/guides/getting-started/api-keys>
- Row Level Security: <https://supabase.com/docs/guides/database/postgres/row-level-security>
- Secure data/API: <https://supabase.com/docs/guides/database/secure-data>
- Storage access control: <https://supabase.com/docs/guides/storage/security/access-control>
- Realtime: <https://supabase.com/docs/guides/realtime>
- Auth: <https://supabase.com/docs/guides/auth>

- RBAC/custom claims: <https://supabase.com/docs/guides/api/custom-claims-and-role-based-access-control-rbac>
- Local migrations: <https://supabase.com/docs/guides/local-development/overview>
- Edge Function secrets: <https://supabase.com/docs/guides/functions/secrets>